Neural Computing and Applications (2025) 37:7207–7260
https://doi.org/10.1007/s00521-024-10848-1

ORIGINAL ARTICLE

Supercell thunderstorm algorithm (STA): a nature-inspired metaheuristic algorithm for engineering optimization

Mohamed H. Hassan[1] · Salah Kamel[2]

Received: 1 November 2023 / Accepted: 29 November 2024 / Published online: 1 February 2025

Abstract
In this paper, an optimization algorithm called supercell thunderstorm algorithm (STA) is proposed. STA draws inspiration from the strategies employed by storms, such as spiral motion, tornado formation, and the jet stream. It is a computational algorithm specifically designed to simulate and model the behavior of supercell thunderstorms. These storms are known for their rotating updrafts, strong wind shear, and potential for generating tornadoes. The optimization procedures of the STA algorithm are based on three distinct approaches: exploring a divergent search space using spiral motion, exploiting a convergent search space through tornado formation, and navigating through the search space with the aid of the jet stream. To evaluate the effectiveness of the proposed STA algorithm in achieving optimal solutions for various optimization problems, a series of test sequences were conducted. Initially, the algorithm was tested on a set of 23 well-established functions. Subsequently, the algorithm's performance was assessed on more complex problems, including ten CEC2019 test functions, in the second experimental sequence. Finally, the algorithm was applied to five real-world engineering problems to validate its effectiveness. The experimental results of the STA algorithm were compared to those of contemporary metaheuristic methods. The analysis clearly demonstrates that the developed STA algorithm outperforms other methods in terms of performance.

Keywords Supercell thunderstorm algorithm · Metaheuristics · Global optimization · Optimization problems

1 Introduction

The difficulty of real-world optimization problems has increased, requiring more efficient approaches to finding solutions. Optimization involves determining the most effective strategy for minimizing or maximizing the objective function of a given problem. Numerous researchers have explored various methods for solving these intricate and challenging real-world problems [1]. In the past twenty years, there has been significant academic interest in nature-inspired computation, as nature provides valuable inspiration for designing artificial computing systems that can discover optimal solutions for complex mathematical problems [2].

Evolutionary algorithms (EAs) are widely recognized as highly effective techniques in research. The literature has seen the utilization of various types of EAs, such as differential evolution (DE) [3], genetic algorithm (GA) [4], and evolutionary programming (EP) [5]. These algorithms tackle diverse challenges by incorporating prior knowledge into an evolutionary search process, effectively exploring a solution space filled with potential solutions. Though EAs are unable to achieve the best solution for many problems in spite of the above-mentioned benefits, hence, several academics have combined these techniques with existing technologies to enhance their solutions [6].

Swarm intelligence (SI) represents an alternative approach within the realm of intelligent computing algorithms such as particle swarm optimization (PSO) [7], which mimics the swarm behavior of birds or fish; ant colony optimization (ACO) [8], which imitates the foraging and schooling behavior of ants; and other algorithms,

✉ Salah Kamel
skamel@aswu.edu.eg

Mohamed H. Hassan
mohamedhosnymoee@gmail.com

1 Ministry of Electricity and Renewable Energy, Cairo, Egypt

2 Department of Electrical Engineering, Faculty of Engineering, Aswan University, Aswan 81542, Egypt

including gravitational search technique (GSA) [9], grey wolf optimization algorithm (GWO) [10], artificial bee colony (ABC) [11], moth flame optimization (MFO) [12], whale optimization algorithm (WOA) [13], and ant lion optimizer (ALO) algorithms [14], and many more. Swarm intelligence addresses various challenges by emulating the natural behavior of animals as they seek food and move between locations. This approach is often influenced by the problem's scale and nonlinearity. Despite the widespread application of conventional analytical methods to resolve computational and combinatorial problems, many of these methods struggle to effectively converge on solutions.

Swarm intelligence offers several advantages. For instance, individual entities within a population can enhance their search efficiency as they move from one location to another, while the collective populations in a swarm collectively progress within their respective locations. In evolutionary algorithms (EAs), weaker and less capable populations are phased out and replaced by highly skilled ones. A swarm consistently ventures into uncharted territories within the search domain, facilitating the rapid discovery of globally optimal solutions. Nevertheless, swarm intelligence is not without its downsides. For instance, the coordinated movements of individuals may lead to entrapment in local optima, and the persistent inability of populations to escape from these regions can prematurely halt the exploration process [15].

Nature-inspired techniques have been devised due to their effectiveness in tackling a multitude of problems. There is an opportunity to blend evolutionary approaches with swarm algorithms to create new capabilities for problem-solving. These capabilities harness the benefits provided by swarm intelligence for efficient exploration within the swarm's optimal regions while also leveraging the power of evolutionary methods to continually traverse the search space and avoid local optima. It's worth noting that the No Free Lunch (NFL) theorem stipulates that no single technique can excel on all problems [16]. This drives researchers to propose novel techniques or enhance existing ones to address specific optimization challenges. Each algorithm possesses its unique strengths and weaknesses, and their application to various real-world problems can yield enhanced outcomes.

Thus, the introduction of techniques such as the supercell thunderstorm algorithm (STA), which is a nature-inspired method for find the optimum solution for the optimization problems, mimics behavior of supercell thunderstorms, which are a type of severe thunderstorm characterized by rotating updrafts, powerful wind shear, and the potential for producing tornadoes. The optimization techniques employed by the STA method are delineated through three distinct approaches: navigating within a dispersed search space via spiral motion, capitalizing on opportunities within a converged search space through tornado formation, and swiftly progressing with a jet stream. To validate the robustness and effectiveness of STA, a comprehensive evaluation is conducted using a combination of twenty-three classical and ten CEC2019 test functions. Moreover, the study employs five engineering design problems to further evaluate the efficiency of the STA attaining optimal solutions for real-world challenges. It is compared with eleven existing techniques from the literature. The findings reveal that STA performs exceptionally, showcasing benefits like minimal time complexity, rapid convergence, high precision in solutions, general applicability, and robustness. In summary, the primary contributions of this research can be outlined as follows:

- Introduction of STA algorithm: This paper introduces the supercell thunderstorm algorithm (STA), an approach designed for tackling global optimization problems and engineering design problems.
- Comprehensive testing: The paper conducts rigorous testing of the proposed STA algorithm across a variety of optimization scenarios. This includes the evaluation of STA on 23 established test functions, the IEEE CEC2019 test suite, and 7 distinct engineering design issues.
- Comparison with contemporary metaheuristic algorithms: The paper performs comparative analysis by benchmarking STA against recent metaheuristic algorithms (MAs). This allows for the assessment of STA's performance relative to other state-of-the-art optimization techniques.
- Enhanced reliability: The experimental results presented in the paper demonstrate that the STA algorithm consistently outperforms alternative optimization algorithms. This substantiates the claim that STA offers a more reliable and effective solution for addressing a wide range of optimization challenges.

The subsequent sections of this article are structured as follows: In Sect. 2, it is provided a brief exploration of the dynamic traits exhibited by storms and expound upon each stage of the newly devised supercell thunderstorm algorithm (STA). Section 3 delineates the evaluation methodology and the outcomes obtained. In Sect. 4, the STA technique is used to solve five real-world engineering design problems. Finally, Sect. 5 encapsulates the conclusions drawn from this research.

2 The supercell thunderstorm algorithm (STA)

In this section, the suggested nature-inspired algorithm, named supercell thunderstorm algorithm (STA), is presented as follows.

2.1 Behavior of supercell thunderstorm

The term "supercell" was coined in the mid-1970s, although these unique convective storms were first identified as a distinct category in the early 1960s. Supercells are the rarest type of storms but are known for generating an exceptionally high amount of severe weather, including large hail, destructive winds, intense lightning, and tornadoes [17]. Almost all instances of hail measuring 5 cm or larger in diameter are associated with supercell storms. Similarly, most of the powerful and violent tornadoes originate from supercells. Supercells are also known for producing some of the highest lightning flash rates ever recorded, with rates exceeding 200 flashes per minute being possible [17]. A supercell is characterized by the presence of a mesocyclone, which is a deep and persistently rotating updraft. Consequently, these storms are sometimes referred to as rotating thunderstorms [18]. Among the four main categories of thunderstorms (supercell, squall line, multicell, and single-cell), supercells are the least common but have the potential to be the most severe. Figure 1 illustrates the behavior of a supercell thunderstorm. The interaction between a strong updraft and highly sheared environmental winds results in a type of storm that outwardly appears distinct from a multicell storm, although the underlying physical processes are similar. Unlike multicell storms, supercells are characterized by a robust and long-lasting rotating updraft.

2.2 STA algorithm

The STA technique mimics the behavior of supercell thunderstorm. Consequently, this algorithm can be separated into three parts, namely, spiral motion, formation, and jet stream. Similar to the majority of metaheuristics, STA operates as a population-based technique. It begins by uniformly distributing the initial solutions across the search space during the initial trial:

$$X_i = \mathrm{lb} + r_1(\mathrm{ub} - \mathrm{lb}),\ i = 1, 2, \ldots, n \tag{1}$$

where lb and ub are the lower and upper bounds of the search space, respectively. r_1 denotes a random vector, which is uniformly distributed in the interval [0, 1]; n is the number of storm populations.

2.2.1 Spiral motion

Supercell thunderstorms are known for their well-defined rotation, often taking on a mesocyclone structure. This rotation is similar to the spiraling motion observed in natural phenomena, such as hurricanes and tornadoes [19]. Figure 2 displays the spiral motion of supercell thunderstorms. The STA seeks to replicate this spiral motion by using mathematical models [1] and algorithms to simulate the development and persistence of rotation within the storm.

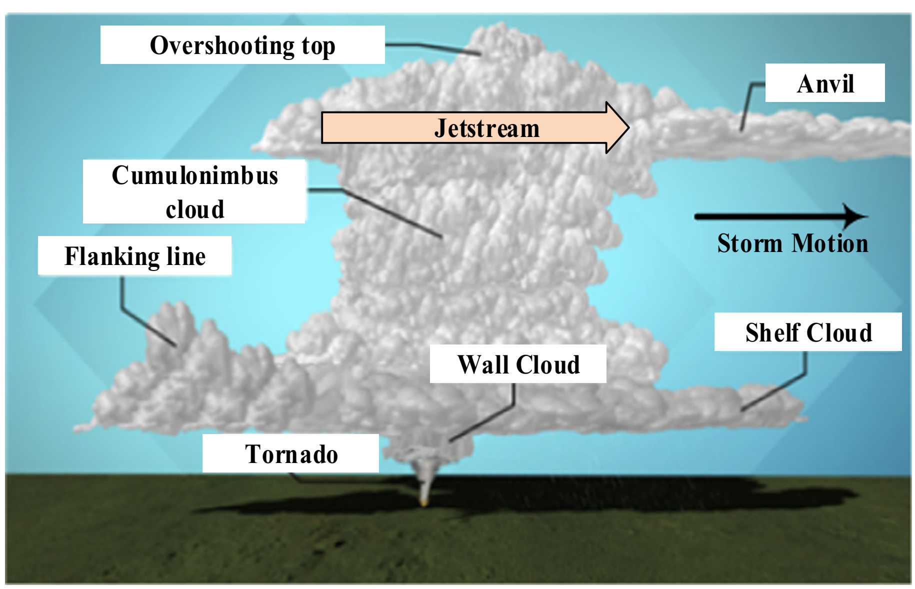


Fig. 1 Behavior of supercell thunderstorm

Fig. 2 Spiral motion of supercell thunderstorms

The mathematical expression for this approach is as follows:

$$
\begin{aligned}
X_i(t+1) &= \begin{cases} \alpha_1.(X_{\text{best}}(t) + \beta_1.|X_{\text{best}}(t) - X_i(t)|) + \alpha_2.X_i(t), & \text{if } r_1 < \dfrac{t}{t_{\max}} \\ \alpha_1.(X_{\text{best}}(t) + \beta_2.|X_{\text{best}}(t) - MG_i(t)|) + \alpha_2.X_i(t), & \text{if } r_1 \geq \dfrac{t}{t_{\max}} \end{cases} \\
\alpha_1 &= a + (1-a).\frac{t}{t_{\max}}, \\
\alpha_2 &= (1-a) - (1-a).\frac{t}{t_{\max}}, \\
\beta_1 &= e^{cs}.\cos(2\pi c), \\
\beta_2 &= e^{cs}.\sin(2\pi c), \\
s &= e^3 + (1-a).\frac{t}{t_{\max}},
\end{aligned}
\tag{2}
$$

In this equation, α_1 and α_2 denote the weight coefficients, which regulate the degree to which individuals move toward the optimum individual and the preceding individual, respectively, while a denotes the control coefficient. The control coefficient is a random value that varies from 0 to 1 through the increase in iteration. The constant controls how far the storms can follow the optimum and preceding individuals throughout the initial stage. t represents the current iteration, and $t_{\max}$ is the maximum number of iterations. $\text{MG}_i(t)$ represents the mean group that is attained using considering the means of a group of points chosen randomly close to the considered solution candidate. Finally, c represents a random number that follows a uniform distribution in the interval [0, 1].

2.2.2 Tornado formation

One of the most destructive aspects of supercell thunderstorms is their ability to produce tornadoes. Tornadoes are highly localized, intense, and violent wind phenomena that form within the rotating updraft of a supercell [20]. STA aims to understand the conditions and processes that lead to tornado formation within supercells and to simulate these processes in its algorithms. By doing so, it helps meteorologists better predict when and where tornadoes may develop within a storm.

The tornado formation has been considered as they have significant impacts on the supercell thunderstorms. The position of the populations is modified based on the tornado formation to avoid falling them into local optimum solutions. This stage can be mathematically described as below [21]:

$$
\begin{aligned}
X_i(t+1) &= \begin{cases} X_i(t) + \left[\text{rand}(0,1) \times \exp\left(-G_1.\dfrac{t}{t_{\max}}\right)\right](X_{r3}(t) - X_{r4}(t)) \\ X_i(t) + [\text{TF }(1 - \text{rand}(0,1)) + \text{rand}(0,1)](X_{r5}(t) - X_{r6}(t)) \end{cases} \\
G_2 &= 2 \times n \times \left(0.1 - 0.05.\frac{t}{t_{\max}}\right)
\end{aligned}
\tag{3}
$$

where the TF equals 0.7. G_1 is the step factor. The symbol r_2 represents a random number while r_3, r_4, r_5, r_6 are the random indices of the storm.

2.2.3 Jet stream

The jet stream is a high-speed, narrow air current located in the upper levels of the Earth's atmosphere. It plays a crucial role in weather patterns and can influence the

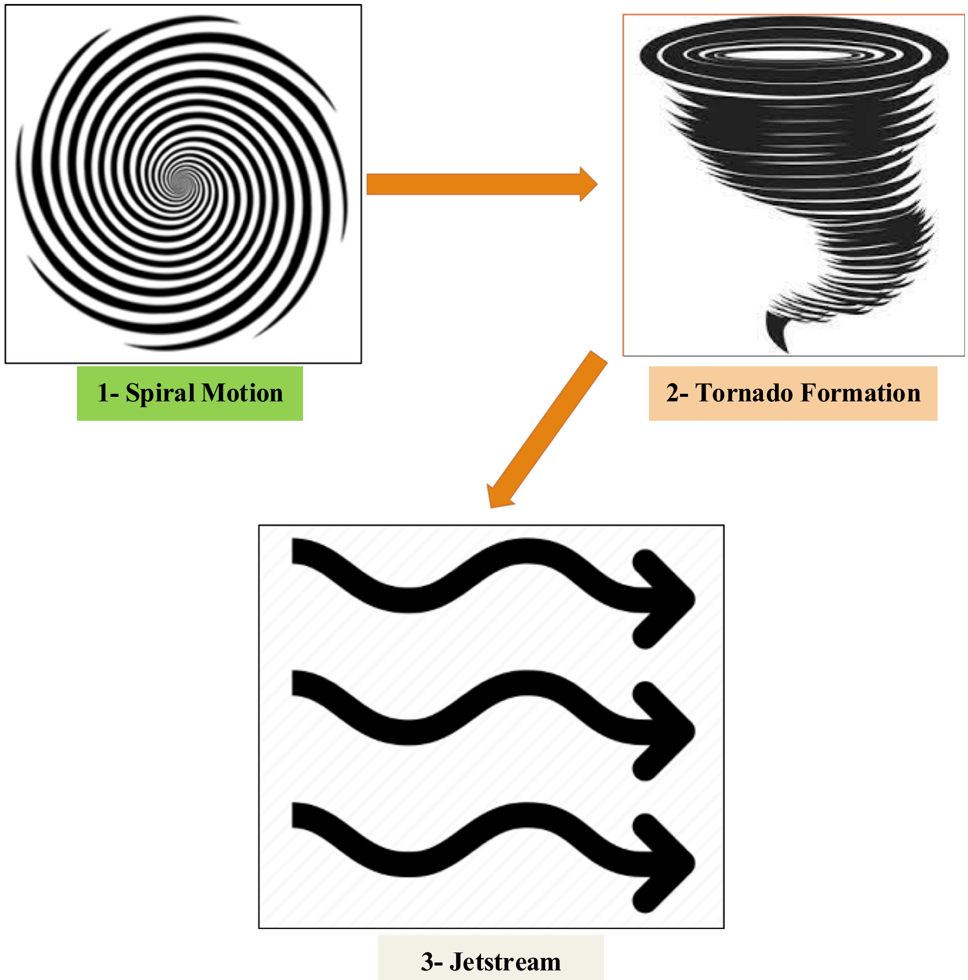


Fig. 3 Co-sequences for the three main stages of supercell thunderstorm by STA

development and movement of severe thunderstorms, including supercells [22]. STA takes into account the influence of the jet stream on supercell thunderstorms, as its interaction with the storm can have a significant impact on their formation, evolution, and track. During this phase, storms prepare to launch an attack on the optimal point within the search area, with all other points converging toward this point. This process can be mathematically expressed as follows [2]:

The movement of the storms takes unlike forms. Within this article, a polar equation is harnessed to graphically represent the trajectory of these storms during Jetstream motion. Additionally, the optimal point is determined by calculating the product of the variance between the current and mean points along the polar x-axis and the variance between the current and optimal points along the polar y-axis.

$$X_i(t+1) = \begin{cases} \text{rand} \times X_{\text{best}}(t) + d_1(i) \times (X_i(t) - 3 \times \text{rand} \times X_{\text{mean}}(t)) + y_1(i) \times (x_i - 2 \times X_{\text{best}}(t)) & r_7 \le 0.3 \\ X_i(t) + \text{rand} \times [X_{\text{best}}(t) - |X_k(t)|] + rand \times [|X_i(t)| - X_k(t)] & r_7 > 0.3 \end{cases}$$

$$d_1(i) = \frac{dr(i)}{\max(|dr|)}, y_1(i) = \frac{yr(i)}{\max(|yr|)} \tag{4}$$

$$dr(i) = r(i) \times \sinh(\theta(i)), yr(i) = r(i) \times \cosh(\theta(i))$$

$$\theta(i) = a_2 \times \pi \times \exp(\text{rand}(\text{rand}(n,1), andr(i) = \theta(i)$$

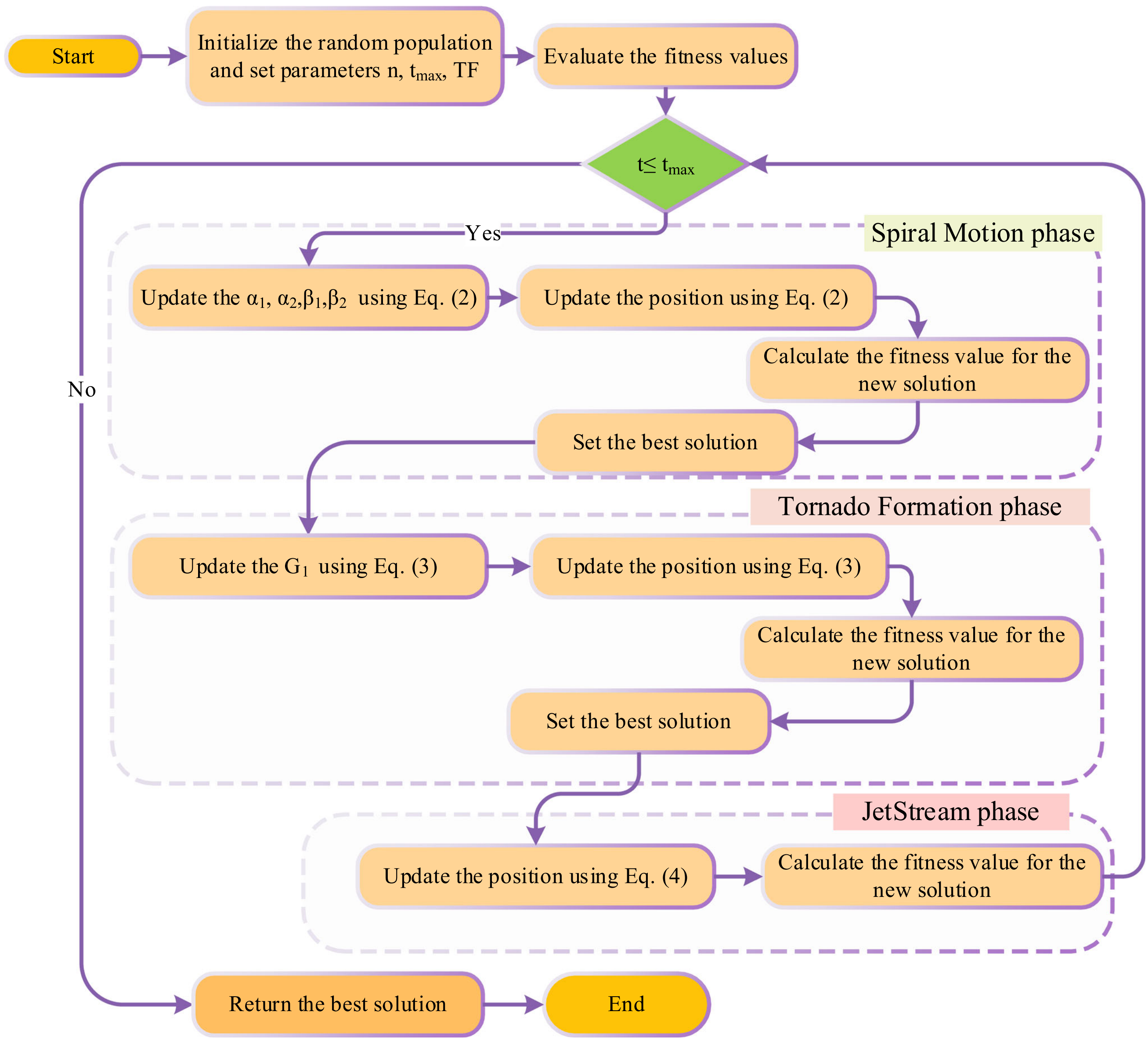


Fig. 4 Flowchart of the STA algorithm

2.3 STA stages, exploration, and exploitation

Figure 3 shows the three stages of the proposed STA algorithm. In the initial optimization phase, as illustrated in stage 1 of the figure, the storms engage in a spiral motion. This spiral motion aids the storms in distinctively exploring their respective areas, leading to a thorough examination of the field. Subsequently, each storm, with its new location, undergoes objective assessment, and if the new location proves more suitable than the previous one, it is adopted. This marks the transition to the second phase of the technique, which aims to facilitate the shift from exploration to exploitation. In this stage, the storms also commence the formation of tornado groups.

The formation of a tornado serves as a pivotal reference point, bolstering the strength of the storms. During this stage, half of the individuals are designated for exploration, while the remaining half take charge of exploitation. As the storm positions draw nearer to each other and the step length decreases compared to the previous stage, the Jetstream plays a crucial role in aiding the proposed STA technique to effectively steer clear of local optima, resulting in improved overall performance. The adaptive convergence factor introduced in this stage plays a significant role in guiding the storms to confine their search efforts to a specific neighborhood for exploitation, preventing them from squandering resources on less promising areas within the domain. The figure outlines the final stage, labeled as stage 3. The flowchart illustrating the STA technique is shown in Fig. 4. Additionally, Algorithm 1 provides the pseudocode defining the STA technique.

Algorithm 1 Pseudocode of the STA technique

```
% STA setting
Set the control parameter (Dimension of problem (d), maximum number of iterations, search agents),
and parameter TF
% Initialization
    Initialize the population randomly
    Assess the objective function of the new solution
   Attain the optimal solution
% Main Loop
     While t ≤ t_max
          Update the α1, α2, β1 and β2 using Eq. (2).
% Spiral Motion stage
         For each population
                Update the location of by Equation (2).
        End For
% Create group
        Assess the objective function of each individual
       Obtain the best solution
% Tornado Formation stage
          Update the G1 using Equation (3).
         For each population
                Update the location of by Equation (3).
        End For
% Create group
        Check the limits of the new positions Assess the objective function of each individual
        Obtain the best solution
 % Jetstream phase
         For each population
                Update the position of using Equation (4).
        End For
% Create group
        Check the limits of the new positions Assess the objective function of each individual
        Obtain the optimal solution
       t=t+1
    End while
   Output the best solution
```

Table 1 Benchmark functions

Function	Dim	Range	$f_{\min}$
$F_1(x) = \sum_{i=1}^{N} x_i^2$	50, 100, 500	[− 100, 100]	0
$F_2(x) = \sum_{i=1}^{N} \lvert x_i \rvert + \prod_{i=1}^{N} \lvert x_i \rvert$	50, 100, 500	[− 10, 10]	0
$F_3(x) = \sum_{i=1}^{N} \left(\sum_{j-1}^{i} x_j \right)^2$	50, 100, 500	[− 100, 100]	0
$F_4(x) = max_i\{\lvert x_i \rvert, 1 \le i \le N\}$	50, 100, 500	[− 100, 100]	0
$F_5(x) = \sum_{i=1}^{N-1} [100\left(x_{i+1} - x_i^2\right)^2 + (x_i - 1)^2$	50, 100, 500	[− 30, 30]	0
$F_6(x) = \sum_{i=1}^{N} (\lvert x_i + 0.5 \rvert)^2$	50, 100, 500	[− 100, 100]	0
$F_7(x) = \sum_{i=1}^{N} i x_i^4 + random[0,1]$	50, 100, 500	[− 1.28, 1.28]	0
$F_8(x) = \sum_{i=1}^{N} -x_i \sin\left(\sqrt{\lvert x_i \rvert}\right)$	50, 100, 500	[− 500, 500]	− 418.9829 × dim
$F_9(x) = \sum_{i=1}^{N} [x_i^2 - 10\cos(2\pi x_i) + 10]$	50, 100, 500	[− 5.12, 5.12]	0
$F_{10}(x) = -20\exp\left(-0.2\sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}\right) - \exp\left(\frac{1}{N}\sum_{i=1}^{N}\cos(2\pi x_i)\right) + 20 + e$	50, 100, 500	[− 32, 32]	0
$F_{11}(x) = \frac{1}{4000}\sum_{i=1}^{N} x_i^2 - \prod_{i=1}^{N}\cos\left(\frac{x_i}{\sqrt{i}}\right) + 1$	50, 100, 500	[− 600, 600]	0
$F_{12}(x) = \frac{\pi}{n}\left\{10\sin(\pi y_1) + \sum_{i=1}^{n-1}(y_i - 1)^2\left[1 + 10\sin^2(\pi y_{i+1})\right] + (y_n - 1)^2\right\} + \sum_{i=1}^{n} u(x_i, 10, 100, 4)$			
$y_i = 1 + \frac{x_i + 1}{4} u(x_i, a, k, m) = \begin{cases} k(x_i - a)^m x_i > a \\ 0 - a < x_i < a \\ k(-x_i - a)^m x_i < -a \end{cases}$	50, 100, 500	[− 50, 50]	0
$F_{13}(x) = 0.1\left\{\sin^2(3\pi x_1) + \sum_{i=1}^{n}(x_i - 1)^2\left[1 + \sin^2(\pi x_i + 1)\right]\right.$	50, 100, 500	[− 50, 50]	0
$\left. + (x_n - 1)^2\left[1 + \sin^2(2\pi x_n)\right]\right\} + \sum_{i=1}^{n} u(x_i, 5, 100, 4)$			
$F_{14}(x) = \left(\frac{1}{500} + \sum_{j=1}^{25}\frac{1}{j + \sum_{i=1}^{2}\left(x_i - a_{ij}\right)^6}\right)^{-1}$	2	[− 65, 65]	1
$F_{15}(x) = \sum_{i=1}^{11}\left[a_i - \frac{x_1\left(b_i^2 + b_i x_2\right)}{b_i^2 + b_i x_3 + x_4}\right]^2$	4	[− 5, 5]	0.00030
$F_{16}(x) = 4x_1^2 - 2.1x_1^4 + \frac{1}{3}x_1^6 + x_1 x_2 - 4x_2^2 + 4x_2^4$	2	[− 5, 5]	− 1.0316
$F_{17}(x) = \left(x_2 - \frac{5.1}{4\pi^2}x_1^2 + \frac{5}{\pi}x_1 - 6\right) + 10\left(1 - \frac{1}{8\pi}\right)\cos x_1 + 10$	2	[− 5, 5]	0.398
$F_{18}(x) = \left[1 + (x_1 + x_2 + 1)^2\left(19 - 14x_1 + 3x_1^2 - 14x_2 + 6x_1 x_2 + 3x_2^2\right)\right]$	2	[− 2, 2]	3
$\times\left[30 + (2x_1 - 3x_2)^2 \times \left(18 - 32x_1 + 12x_1^2 + 48x_2 - 36x_1 x_2 + 27x_2^2\right)\right]$			
	3	[1, 3]	− 3.86

Table 1 (continued)

Function	Dim	Range	$f_{\min}$
$F_{19}(x) = -\sum_{i=1}^{4} c_i exp\left(-\sum_{j=1}^{3} a_{ij}(x_j - p_{ij})^2\right)$	6	[0, 1]	− 3.32
$F_{20}(x) = -\sum_{i=1}^{4} c_i exp\left(-\sum_{j=1}^{6} a_{ij}(x_j - p_{ij})^2\right)$	4	[0, 10]	− 10.1532
$F_{21}(x) = -\sum_{i=1}^{5} \left[(X - a_i)(X - a_i)^T + c_i\right]^{-1}$	4	[0, 10]	− 10.4028
$F_{22}(x) = -\sum_{i=1}^{7} \left[(X - a_i)(X - a_i)^T + c_i\right]^{-1}$	4	[0, 10]	− 10.5363
$F_{23}(x) = -\sum_{i=1}^{10} \left[(X - a_i)(X - a_i)^T + c_i\right]^{-1}$			

2.3.1 Computational complexity analysis of the proposed STA algorithm

Computational complexity serves as an important metric for assessing the effectiveness of methods in dealing with optimization problems. Numerous factors impact a technique's complexity, including the number of individuals involved (n), the dimensionality of the problem's variables (d), and the maximum number of iterations ($t_{\max}$). In the case of STA, the comprehensive computational complexity can be defined as follows:

$$
\begin{aligned}
O(\mathrm{STA}) &= O(\text{problem definition}) + O(\text{initialization}) \\
&\quad + O(\text{function evaluation}) + O(\text{position updating in Spiral Motion phase}) \\
&\quad + O(\text{position updating in Tornado Formation phase}) \\
&\quad + (\text{position updating in Jetstream phase}) \\
&= O\left(1 + n + Tn + \frac{1}{2}Tnd + \frac{1}{2}Tnd + Tnd\right) \cong O(2Tnd + Tn + n)
\end{aligned}
$$

3 Experimental results and discussion

The existing examination evaluates the efficiency of the STA by several benchmark functions and real-world optimization issues. Two series of benchmark test functions ensemble including twenty-three classical functions, and ten CEC2019 benchmarks are employed to the numerical assessment phase; furthermore, five engineering optimization problems have been selected as exemplars of real-world challenges. The STA algorithm has been employed for 500 iterations with a population size of 50 to address both the benchmark functions and practical applications. To assess the stability and reliability of the STA algorithm, it has been implemented independently 30 individual runs. The results, including the mean performance, standard deviation (STD), the worst-so-far solutions, and the best-so-far solutions, have been documented. In order to affirm the superior performance of STA, it has been subjected to a rigorous comparison with a renowned metaheuristic optimization technique using both the Wilcoxon rank-sum test (WRST) and Friedman's mean rank test. These tests serve to validate the STA algorithm's dominance, as detailed in the subsequent sections.

3.1 Definition of twenty-three classical test functions

In this subsection, a set of widely recognized benchmark functions is used to assess the performance of the STA method introduced in this paper. These benchmark functions are categorized into three groups: unimodal, multimodal, and multimodal functions with fixed dimensions. The group of unimodal benchmark test functions includes

Fig. 5 Qualitative metrics of some mathematical benchmark functions: 2D views of the functions, search history, and convergence curve using the STA algorithm

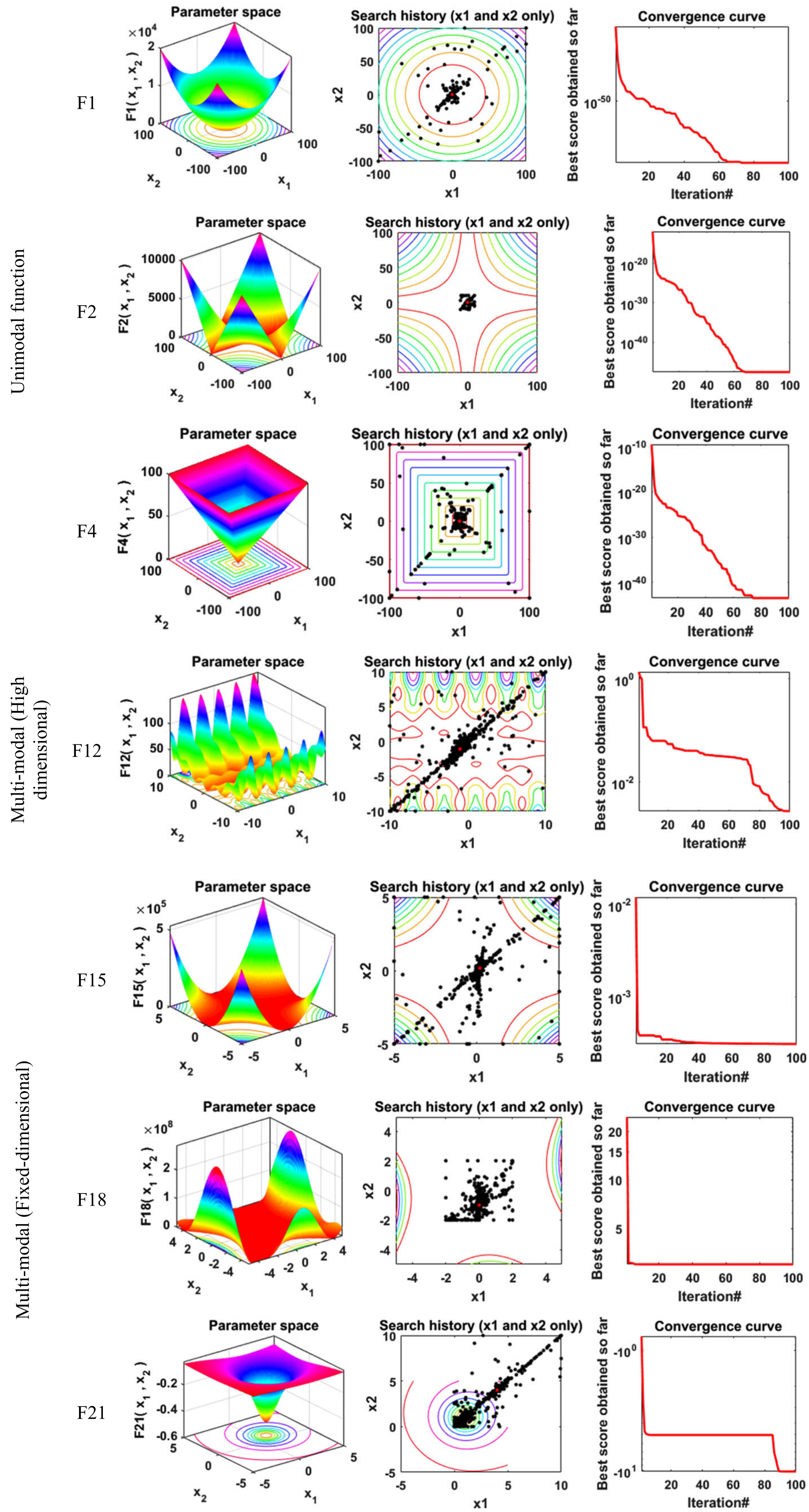


Table 2 Scenarios of the tuning parameter

Scenario no	TF value
1	0.1
2	0.2
3	0.3
4	0.4
5	0.5
6	0.6
7	0.7
8	0.8
9	0.9

seven functions denoted as F1 to F7. Each of these functions has a single global optimal solution, making them a common choice for evaluating the algorithm's proficiency in effectively exploiting local solutions. On the other hand, the multimodal functions, from F8 to F23, not only feature a globally optimal solution but also comprise multiple locally optimal solutions. These functions are used to challenge the technique's ability to explore globally and avoid falling into local optima. Table 1 presents the mathematical formulas and characteristics of these benchmark functions.

3.2 Convergence analysis of STA

Figure 5 shows qualitative metrics for the analysis of the convergence and performance of the proposed STA technique in the benchmark functions. The first column of the chart provides a visual representation of the functions in a two-dimensional format, offering vision into the topological characteristics of the search space. The second column depicts the search history, serving as the first discussed metric. This representation showcases a more clustered distribution of the population around optimal points in unimodal functions and a more dispersed performance in the case of multimodal functions. The former pattern aids in effectively exploiting results as needed for unimodal functions, while the latter signifies exploration across the domain, assisting the proposed STA algorithm in searching the entire space for multimodal functions.

The third metric is the convergence curve, illustrating the progress of the best solution found up to that point. The figure presents a definite shape for the convergence curve corresponding to each function group. In unimodal functions, this curve appears notably smooth, showcasing a consistent improvement in results as iterations progress. Conversely, for multimodal functions, the curve takes on a stepwise behavior, a characteristic often observed in such functions. It's evident from each group that in unimodal functions, STA quickly identifies and converges toward the optimal solution in the early iterations, striving to enhance solutions as iterations continue. In contrast, for multimodal functions, populations persistently explore the entire area even in the later iterations, in their pursuit of discovering better solutions.

3.3 Sensitive analysis of the STA

This subsection emphases on the sensitivity examination of a control parameter utilized in STA. This parameter is TF that contribute to the optimization process. This parameter is evaluated at one value from 0.1 to 0.9; consequently, there are 9 scenarios for each value of TF parameter as shown in Table 2. In this evaluation, The STA technique is employed for 500 iterations with a population size of 50 to address eight benchmark functions (namely, F1, F3, F5, F8, F10, F13, F17, and F21).

Table 3 displays the statistical outcomes achieved across these functions in different scenarios. An analysis of the results reveals that the seventh scenario, where TF (Transfer function) is set to 0.7, consistently produces superior results across all tested functions when compared to other scenarios. Subsequently, the sixth and eighth scenarios secure the second and third positions, respectively, in terms of performance.

3.4 Scalability analysis of the STA

The effect of varying the number of solutions is studied across a range of benchmark functions. To comprehensively assess STA's sensitivity to parameters, a multitude of solution numbers (i.e., 10, 20, 30, 40, and 50) are systematically tested, allowing for a comparison of the changes in solution numbers over 500 iterations. The results demonstrate that even when employing different search agents, the STA optimizer retains its advantages, indicating that STA exhibits robustness and is less susceptible to the influence of population size variations. Also, the results show that the best solution for the most functions when the population size is 40 as shown in Fig. 6. Therefore, the population size will be 40 for the rest of problem in the whole of paper.

3.5 Simultaneous analysis of population size and transfer function parameter

To confirm the choice of a transfer function (TF) value of 0.7 and to understand the interplay between population size and TF parameters, it is essential to conduct a thorough analysis. These factors are often interdependent, meaning that changes in one can significantly influence the behavior and outcomes of the other. This section explores the necessity and benefits of simultaneously analyzing population size and TF parameters to achieve more precise and

Table 3 Effect of the STA parameter (i.e., TF) tested on several benchmark functions

Fun no	Measure	Scenario no								
		Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Scenario 6	Scenario 7	Scenario 8	Scenario 9
F1	Best	5.1E-285	4.1E-285	2.1E-293	2.5E-297	0	0	0	0	0
	Mean	2E-252	8.7E-255	9.2E-259	5.4E-263	3.8E-272	8.4E-281	1.5E-298	0	0
	Worst	4E-251	1.7E-253	1.4E-257	1.1E-261	7.5E-271	1.5E-279	3E-297	0	0
	Std	0	0	0	0	0	0	0	0	0
F3	Best	9E-284	1.1E-286	1.6E-293	5.7E-296	0	0	0	0	0
	Mean	6.2E-247	1.1E-244	1.9E-254	6E-259	1.7E-253	6.1E-291	3.2E-302	0	0
	Worst	1.2E-245	2.2E-243	3.7E-253	1.2E-257	3.4E-252	1.1E-289	6.4E-301	0	0
	Std	0	0	0	0	0	0	0	0	0
F5	Best	0.031441	0.052195	0.133658	0.078908	0.074772	0.009377	0.487775	0.156419	0.196188
	Mean	8.46689	10.68305	21.10775	13.33364	15.86999	5.617916	14.07777	5.762929	12.74664
	Worst	26.78626	26.75084	27.06626	26.82969	26.84433	26.71339	26.68714	26.49116	26.80643
	Std	12.21611	13.42055	11.04624	13.78411	13.39902	11.03442	12.998	10.82241	13.03734
F8	Best	− 12,568.6	− 12,569.4	− 12,569.1	− 12,568.6	− 12,569.2	− 12,569.4	− 12,569.5	− 12,569.2	− 12,568.6
	Mean	− 12,546.5	− 12,568	− 12,563.9	− 12,486.3	− 12,519.9	− 12,567.7	− 12,568.6	− 12,521.8	− 12,438.2
	Worst	− 12,350.1	− 12,566.7	− 12,527.6	− 12,222.3	− 12,220.8	− 12,565.2	− 12,567.8	− 12,349.3	− 11,891.2
	Std	69.01128	0.800318	12.78822	129.2528	112.8739	1.515989	0.484987	91.10059	215.8892
F10	Best	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16
	Mean	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16
	Worst	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16	8.88E-16
	Std	0	0	0	0	0	0	0	0	0
F13	Best	0.000372	0.001322	0.000881	0.000442	0.000745	0.001451	0.001849	0.001924	0.003582
	Mean	0.004026	0.006679	0.008521	0.00815	0.007496	0.004951	0.011466	0.024497	0.011765
	Worst	0.012912	0.016034	0.023166	0.016596	0.024839	0.014294	0.030762	0.134355	0.026751
	Std	0.004762	0.005972	0.008577	0.006607	0.008062	0.004975	0.011046	0.040342	0.008792
F14	Best	0.998004	0.998004	0.998004	0.998004	0.998004	0.998004	0.998004	0.998004	0.998004
	Mean	1.987311	1.196809	1.295817	2.284728	1.79204	1.196414	1.693032	2.768162	2.384131
	Worst	5.928845	1.992031	2.982105	5.928845	2.982105	2.982105	2.982105	10.76318	5.928845
	Std	1.612014	0.419119	0.669811	1.616004	0.911728	0.627428	0.816969	2.977148	1.557577
F17	Best	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887
	Mean	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887
	Worst	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887
	Std	0	0	0	0	0	0	0	0	0

Table 3 (continued)

Fun no	Measure	Scenario no								
		Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Scenario 6	Scenario 7	Scenario 8	Scenario 9
F21	Best	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 10.1532
	Mean	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 9.13816	− 10.146
	Worst	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 10.1532	− 5.0552	− 10.0811
	Std	2.68E-08	3.69E-09	2E-08	1.43E-08	9.28E-09	2.68E-08	1.74E-09	2.139925	0.022797
Mean	5.422222	5.238889	5.572222	5.661111	5.088889	4.172222	4.166667	4.75	4.927778	
Ranking	7	6	8	9	5	2	1	3	4	

reliable results. Specifically, three scenarios for the TF parameter values (0.6, 0.7, and 0.8) are considered, with a constant population size of 40. From these scenarios, Scenario 7a (TF = 0.7 and population size = 40) achieved the best solutions for the most functions. This indicates that a TF value of 0.7, in combination with a population size of 40, provides a more optimized and effective modeling framework as shown in Table 4 and Fig. 7.

3.6 Dynamic evolution of other parameters in STA algorithm

This subsection illustrates the dynamic changes in the other parameters over iterations. The resulting plots, displayed in Fig. 8a–c, demonstrate how these parameters evolve throughout the optimization process, providing visual insight into their behavior.

3.7 Performance of the proposed STA algorithm

In this subsection, we compare the results obtained using the STA algorithm with those of recent optimization algorithms such as the tunicate swarm algorithm (TSA) [23], salp swarm algorithm (SSA) [24], sine cosine algorithm (SCA) [25], multi-verse optimizer (MVO) [26], moth flame optimization (MFO) [12], grey wolf optimizer (GWO) [10], whale optimization algorithm (WOA) [13], Runge–Kutta optimizer (RUN) [27], weighted mean of vectors (INFO) [28], particle swarm optimization (PSO) [29], and differential evolution (DE) [3]. The analysis was conducted using MATLAB R2016a on a Windows 8.1, 64-bit operating system. All computations were performed on a computer with an Intel Core i5-4210U CPU running at a speed of 2.40 GHz and 8 GB of RAM. Table 5 presents the parameter settings for the optimization methods.

3.8 Stability analysis of STA

With the purpose of evaluate the consistency and effectiveness of the proposed STA optimizer in handling high-dimensional optimization challenges, thirteen functions from Tables 6, 7, and 8 have been employed. These functions are tested at three different dimensional levels, specifically, 50, 100, and 500 dimensions. The tables present calculated values for the worst, average, best, and standard deviation (STD) across the benchmark functions, providing a comprehensive overview of the algorithm's performance in these high-dimensional scenarios. These tables present the results of STA and other algorithms on unimodal test functions (F1–F7). The results indicate that STA consistently outperforms the majority of the tested approaches across a significant portion of these evaluation functions. These findings serve to underscore STA's

Fig. 6 Effect of the STA search agents tested on several benchmark functions

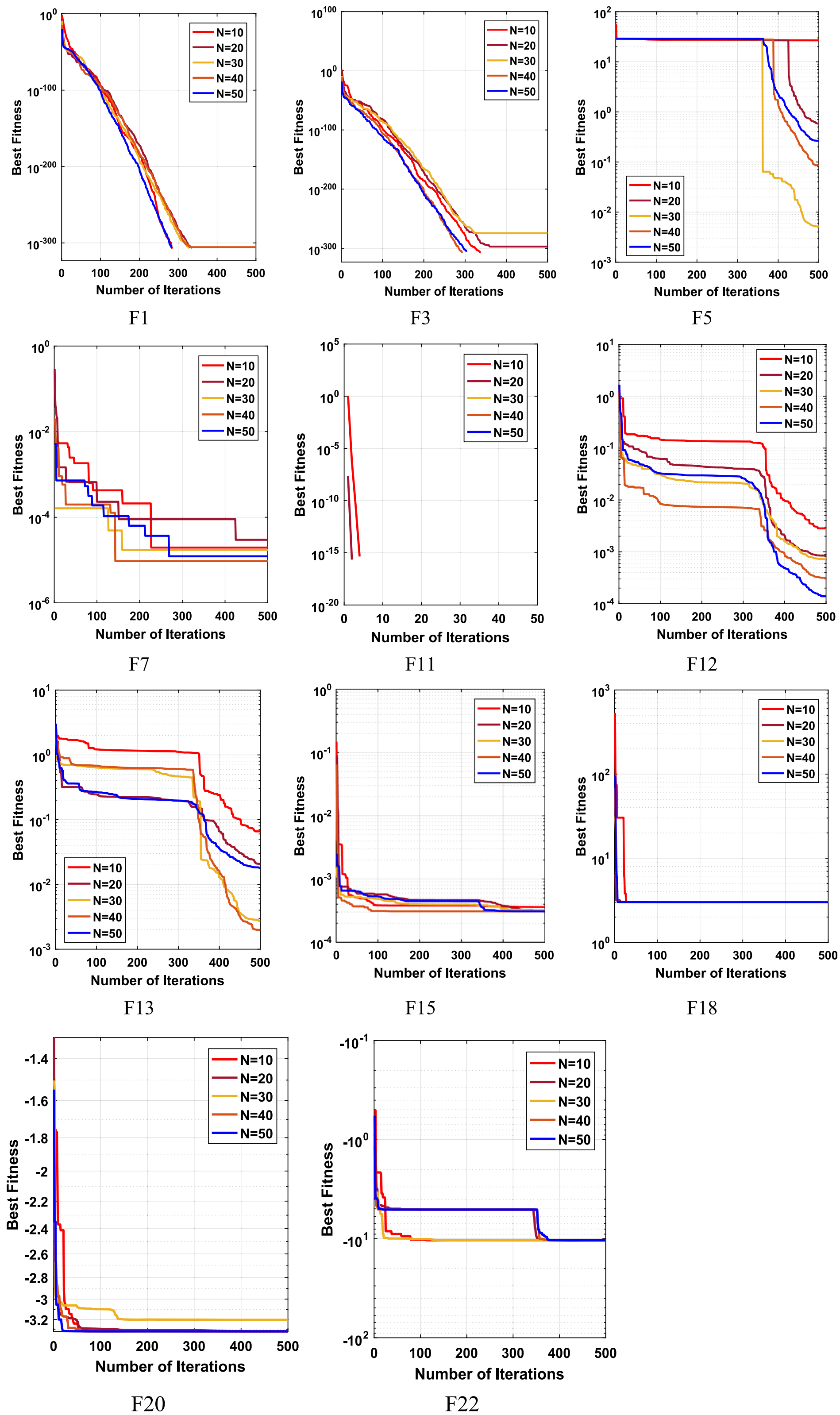


Table 4 Effect of the STA parameters (i.e., TF and population size) tested on several benchmark functions

Fun no	Measure	Scenario no		
		Scenario 6a	Scenario 7a	Scenario 8a
F1	Best	0	0	0
	Mean	6.1E-282	0	4.5E-291
	Worst	6.1E-281	0	4.5E-290
	Std	0	0	0
F3	Best	0	0	0
	Mean	3.5E-279	0	1.5E-291
	Worst	3.5E-278	0	1.5E-290
	Std	0	0	0
F5	Best	0.149669	0.094993	0.255372
	Mean	8.589324	8.16284	10.53579
	Worst	26.80054	27.19263	26.75991
	Std	12.29507	12.58151	12.16273
F8	Best	− 12,568.6	− 12,569.5	− 12,568
	Mean	− 12,539.6	− 12,564.9	− 12,474.3
	Worst	− 12,344	− 12,553.9	− 11,986.9
	Std	69.51059	5.052002	186.7556
F10	Best	8.88E-16	8.88E-16	8.88E-16
	Mean	8.88E-16	8.88E-16	8.88E-16
	Worst	8.88E-16	8.88E-16	8.88E-16
	Std	0	0	0
F13	Best	0.003129	0.001756	0.002429
	Mean	0.016029	0.012463	0.013953
	Worst	0.032028	0.043148	0.03904
	Std	0.010916	0.012456	0.013755
F14	Best	0.998004	0.998004	0.998004
	Mean	3.061278	1.692637	2.570147
	Worst	10.76318	2.982105	10.76318
	Std	3.274854	0.941212	2.990575
F17	Best	0.397887	0.397887	0.397887
	Mean	0.397887	0.397887	0.397887
	Worst	0.397887	0.397887	0.397887
	Std	0	0	0
F21	Best	− 10.1532	− 10.1532	− 10.1532
	Mean	− 10.1532	− 10.1532	− 10.1532
	Worst	− 10.1532	− 10.1532	− 10.1532
	Std	2.13E-06	2.83E-08	7.21E-08
Ranking	2	1	3	

remarkable capacity for exploitation, enabling it to efficiently converge toward optimal solutions with precision and speed. Functions F8 through F13, characterized by their multimodal nature in high dimensions, are specifically scrutinized in the tables. The data presented in these tables clearly illustrate STA's exceptional prowess in exploration when compared to alternative methods. Notably, in the case of high-dimensional multimodal functions, STA surpasses all other algorithms. Furthermore, it's worth noting that STA often achieves the global optimum in most problems with a precision level (Std) that closely rivals that of high-performance algorithms. STA's proficiency in

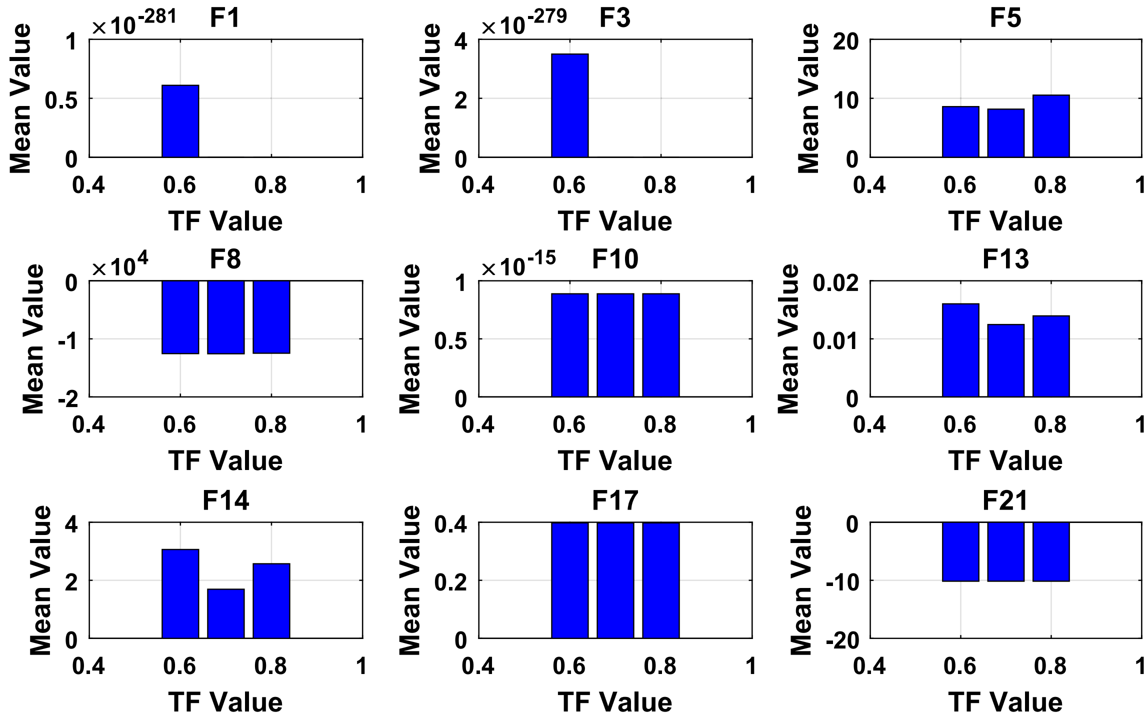


Fig. 7 Simultaneous analysis of population size and transfer function (TF) parameter in STA

exploration is attributed to its unique optimization stages. In summary, the extensive evaluation demonstrates the remarkable proficiency of the proposed STA algorithm in terms of both exploitation and exploration, as observed across a range of benchmark functions. It consistently outperforms other algorithms on the majority of test problems, underscoring its effectiveness in locating global optima for both unimodal and multimodal functions. The innovative jetstream mechanism plays a pivotal role in augmenting the exploration process, enabling the algorithm to navigate the solution space efficiently and uncover high-quality solutions.

The test results of STA when it is used to solve the composite benchmark functions are displayed in Table 9. The means in the table present that STA is superiorly competitive on the composite benchmark functions, performing superlative on eight of the ten functions.

3.8.1 Wilcoxon's rank test results

In this subsection, the differences between STA and other techniques are further analyzed statistically using the Wilcoxon rank-sum test (WRST), which is a paired test that checks for significant differences between two algorithms. The results of the test between STA and each technique at a significance level of α = 0.05 are presented in Tables 10, 11, 12, and 13, where the symbols "+/=/ − " show whether STA executes better, likewise, or worse than the comparison technique. Tables also present the statistical outcomes for STA across various dimensions and functions, indicating whether STA exhibits superior, equivalent, or inferior performance compared to the reference algorithm. Notably, STA consistently surpasses other well-known techniques in the statistical assessments of F1–F13 across different dimensions (Dim=50, 100, 500) and in fixed-dimensional functions F14–F23. These results affirm the substantial superiority of STA across a wide array of functions when compared to alternative techniques. Therefore, it is concluded that the STA technique shows the best performance compared to well-known optimization techniques.

3.8.2 Friedman's rank test results

Table 14 shows the statistical results attained by Friedman tests for Dim=50 [30]. The smaller the ranking value, the better the performance of the techniques. The statistical results obtained by Friedman tests between STA and each technique are presented in Tables 14, 15, 16, and 17. From the results, the highest ranking presents that STA is the best optimizer among the eight techniques.

Furthermore, radar charts and mean ranks, obtained through the Friedman test for all 23 benchmark functions across various algorithms, are depicted in Figs. 9 and 10. These visual representations make it clear that the proposed STA algorithm achieves the lowest mean rank value,

Fig. 8 Parameter's evolution over 500 iterations, demonstrating its variation as influenced by iteration number and random factors

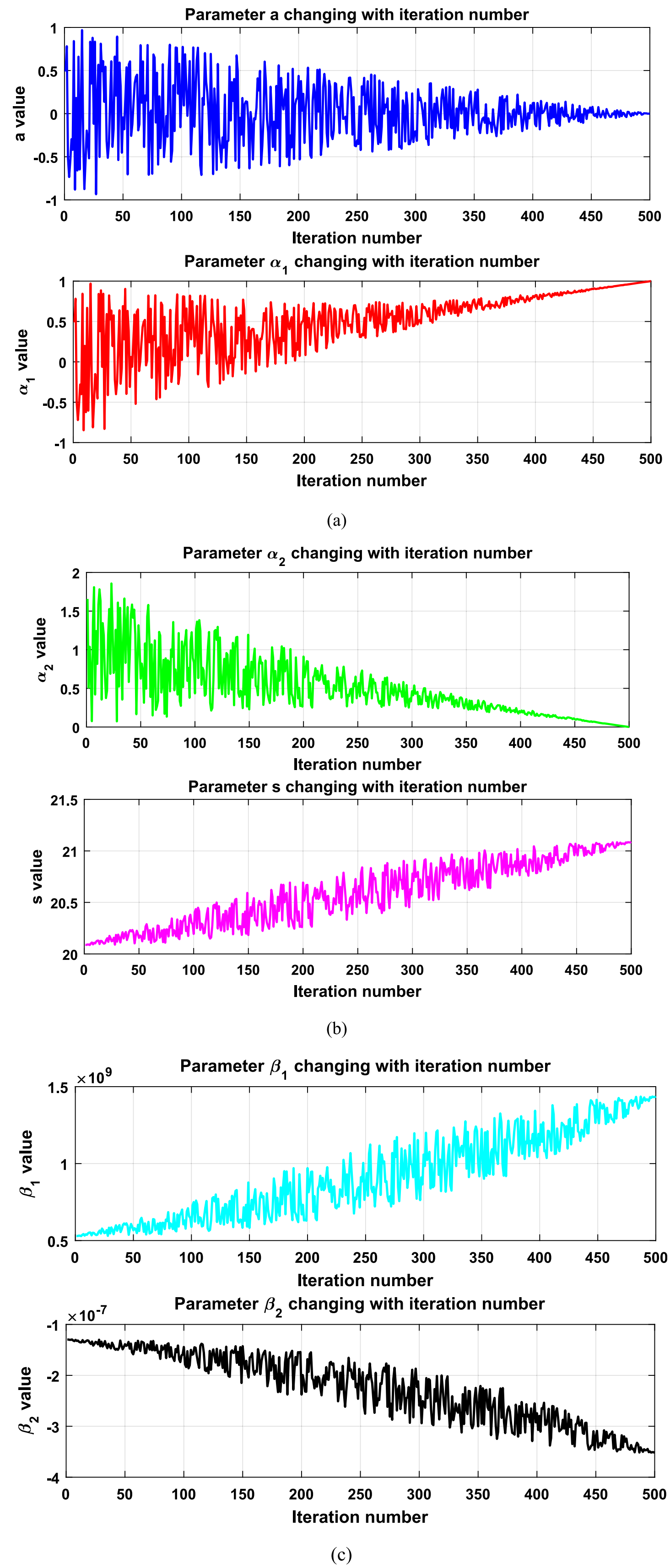


Table 5 Configuration parameters of optimization algorithms used for the comparative evaluation of the STA algorithm

Algorithms	Parameters setting	Value
STA	TF	0.7
TSA	Parameter $p_{\min}$	1
	Parameter $p_{\max}$	4
SSA	v_0	0
SCA	A	2
MVO	Existence probability	[0.2 1]
	Traveling distance rate	[0.6 1]
MFO	Convergence constant a	$[-2\ -1]$
	Spiral factor b	1
GWO	Convergence constant a	[2 0]
WOA	Convergence constant a	[2 0]
	Spiral factor b	1
INFO	Updating rule, vector combining, and a local search	$c = 2$, $d = 4$
PSO	Topology	Fully connected
	Cognitive constant	1
	Social constant	1
	Inertia weight	Linear reduction from 0.9 to 0.1
	Velocity limit	10% of the dimensions range of the variables
DE	Lower bound of scaling factor	0.2
	Upper bound of scaling factor	0.8
	Crossover probability	0.2

signifying its top-ranking status among all the techniques. As a result, according to the Friedman test methodology, the STA technique emerges as the best performing algorithm in this comparative analysis.

3.9 The performance of the STA technique for the CEC2019 function

The algorithm benchmarking made use of the CEC2019 function, and the evaluation of their performance was conducted with the demanding benchmark functions from the CEC2019 test suite, as outlined in Table 18 [31]. This test suite features ten exceptionally challenging and complex composite functions, providing a stringent assessment for the algorithms. The profiles of these benchmark functions are elaborated in Table 18. As described in the previous section, the proposed STA and seven other algorithms for comparison were individually run 20 times on each function, with fixed parameters of a maximum iteration (500) and a population size (20).

The results, including the best, mean, worst values, and standard deviation, derived from this testing, are presented in Table 19. An examination of Table 19 reveals that STA surpasses the other six algorithms in performance across these test functions, with the exception of the SSA algorithm. To sum it up, this section extensively validates the effectiveness and superiority of the STA method through a series of experiments on both classical benchmark functions and the IEEE CEC2019 test suite. Whether dealing with straightforward or intricate numerical challenges, STA consistently delivers commendable results.

The convergence plot of the executed algorithms is depicted in Fig. 11. A close examination of these curves reveals that, for most functions, STA exhibits comparable convergence. Furthermore, the findings indicate that STA attains more stable and resilient solutions across the range of functions tested.

3.9.1 Wilcoxon's rank test results

In this subsection, the differences between STA and other techniques are further analyzed statistically by the Wilcoxon rank-sum test (WRST). Table 20 presents the statistical results of STA for CEC 2019 benchmark functions, signifying whether STA achieves better, similarly, or worse than the comparison technique. STA outperforms other recent techniques in the statistics of CEC 2019 benchmark functions, which approves the significant dominance of STA in most functions compared to other techniques.

Table 6 Numerical results of test functions by the STA technique and other recent techniques (Dim = 50)

Function		STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F1	Best	0	4.94E-18	0.004906	33.32809	5.210607	335.0467	1.04E-22	7.7E-92	3.2E-189	2.22E-54	71.98515	1.646147
	Mean	0	3.64E-17	0.016992	297.8526	6.336375	6730.139	3.94E-22	9.61E-80	9.7E-181	1.47E-53	156.8815	2.264439
	Worst	0	9.95E-17	0.031853	596.2896	9.423383	20,986.97	1.9E-21	7.26E-79	9.7E-180	2.58E-53	220.9806	3.448237
	Std	0	3.32E-17	0.008844	185.8823	1.149383	7003.32	5.39E-22	2.33E-79	0	8.59E-54	48.31066	0.524019
F2	Best	1.1E-166	7.79E-12	3.43301	0.027616	2.545176	25.89061	4.3E-14	8.21E-58	2.6E-106	1.37E-26	7.300573	0.354093
	Mean	2.6E-156	2.73E-11	6.859054	0.273129	211.4024	69.51458	1.43E-13	9.2E-54	9.7E-101	2.92E-26	11.19535	0.438072
	Worst	1.9E-155	7.08E-11	11.18203	0.884014	1026.436	109.9938	3.08E-13	6.52E-53	7E-100	5.08E-26	20.28549	0.546616
	Std	6.2E-156	2.34E-11	2.512395	0.274395	301.739	26.07109	8.36E-14	2.06E-53	2.3E-100	1.04E-26	3.69193	0.067005
F3	Best	0	0.06713	2860.627	19,823.75	2346.106	28,472.9	0.000113	150,280.6	7.3E-172	7.91E-51	3356.175	95,848.25
	Mean	0	3.904113	5372.172	47,214.44	3631.238	66,863.24	0.058813	189,225.4	1E-156	3.14E-50	7534.004	115,156.4
	Worst	2.7E-308	22.01668	13,062.56	71,107.93	5632.253	101,453.1	0.204459	242,694.3	6.9E-156	7.19E-50	13,233.57	145,512.4
	Std	0	6.649753	3029.367	17,275.57	1057.151	22,405.94	0.069756	26,995.59	2.3E-156	2.03E-50	3531.584	15,138.68
F4	Best	1.1E-165	1.132868	11.66573	54.31861	6.682571	71.55181	1.79E-05	0.007274	8.8E-92	9.38E-28	15.52084	43.01534
	Mean	3.9E-147	5.9826	16.01565	68.22212	12.39642	80.77858	9.3E-05	63.79077	2.05E-82	1.92E-27	19.48773	46.30453
	Worst	3.9E-146	14.71519	20.09609	79.46513	18.59179	85.50423	0.000317	89.64472	2.02E-81	3.03E-27	23.74059	50.85756
	Std	1.2E-146	3.560643	2.781428	7.48548	3.894705	4.296418	9E-05	27.11176	6.4E-82	7.22E-28	2.584407	2.655663
F5	Best	0.507384	46.29801	223.015	555,023.3	152.0399	86,127.17	45.88793	47.36974	42.30211	42.01119	1126.573	1584.619
	Mean	2.482297	48.31843	629.1213	4,621,271	1075.32	374,194.9	47.10489	48.28409	45.08219	42.91719	9265.137	2722.944
	Worst	10.05396	48.82544	1467.098	11,707,299	3195.382	1,114,389	48.42923	48.61907	47.71038	45.76272	32,379.57	5385.467
	Std	2.805999	0.789266	488.8727	4,001,865	995.0657	385,811	0.975488	0.39635	1.69276	1.22528	9547.532	1270.351
F6	Best	0.008724	4.790712	0.004036	22.16827	4.513485	694.4274	1.498429	0.227273	1.17E-08	5.19E-07	26.92149	1.585543
	Mean	0.01149	6.459395	0.071517	1018.466	7.001966	8642.816	2.427215	0.5979	1.93E-08	3.95E-06	169.6937	2.480075
	Worst	0.014997	8.165545	0.269641	2431.091	9.977119	19,963.65	3.442145	0.95174	3.81E-08	1.97E-05	327.6728	3.717883
	Std	0.002124	1.173835	0.092692	835.1625	1.640179	5938.867	0.700231	0.227541	8.42E-09	5.88E-06	103.7083	0.670122
F7	Best	2.86E-06	0.009345	0.276106	0.318376	0.045404	1.339389	0.001459	0.0001	3.04E-05	8.91E-05	0.254364	0.152597
	Mean	3.2E-05	0.017204	0.359529	3.627413	0.086992	11.59266	0.002251	0.00459	0.000164	0.001071	0.364667	0.186942
	Worst	9.01E-05	0.030836	0.463542	14.25454	0.108367	47.45529	0.00355	0.015745	0.000266	0.002323	0.531178	0.210389
	Std	2.93E-05	0.006706	0.069661	4.551747	0.018973	13.65194	0.000767	0.00576	7.73E-05	0.000915	0.093351	0.019745
F8	Best	− 20,948.3	− 10,201.3	− 12,471.6	− 6256.98	− 13,669.2	− 16,289.6	− 11,229.8	− 20,949	− 14,251.9	− 14,820	− 11,614.8	− 11,229.8
	Mean	− 20,747.9	− 8903.03	− 11,301.3	− 5054.39	− 12,690.7	− 13,808.3	− 9578.55	− 17,819.2	− 12,525.8	− 13,981.5	− 9848.42	− 9578.55
	Worst	− 19,632.4	− 8046.88	− 10,410.1	− 4526.78	− 10,903.5	− 11,593.3	− 7952.33	− 13,657.4	− 10,966.3	− 12,712.2	− 8050.61	− 7952.33
	Std	409.8983	652.4233	635.3939	567.5683	820.7617	1252.331	877.3302	3213.602	1031.787	639.7093	1040.304	877.3302
F9	Best	0	234.6548	27.89215	33.1459	141.3051	275.3027	2.84E-13	0	0	0	70.14356	227.472
	Mean	0	385.7538	73.24871	99.37777	232.2828	334.1206	1.783417	0	0	0	91.95302	253.1081
	Worst	0	466.2065	98.54029	184.8707	286.3291	439.9644	6.294313	0	0	0	139.4802	271.2363
	Std	0	71.44653	25.66795	46.52476	48.83893	55.24766	2.506596	0	0	0	23.16544	14.55975

Table 6 (continued)

Function		STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F10	Best	8.88E-16	3.96E-10	2.944608	1.941069	2.251562	9.536026	1.29E-12	8.88E-16	8.88E-16	8.88E-16	4.247921	0.448162
	Mean	8.88E-16	0.793999	3.969388	13.88716	2.898352	18.61905	3.76E-12	3.02E-15	8.88E-16	8.88E-16	6.098461	0.554878
	Worst	8.88E-16	3.170612	5.106013	20.47026	4.198885	19.96462	8.14E-12	7.99E-15	8.88E-16	8.88E-16	8.471722	0.746894
	Std	0	1.296854	0.661868	8.48939	0.577439	3.242559	2.15E-12	2.48E-15	0	0	1.199777	0.098698
F11	Best	0	0	0.086347	1.746396	1.043482	3.243613	0	0	0	0	1.629435	0.819291
	Mean	0	0.006021	0.176798	8.855115	1.063598	64.79344	0.004122	0	0	0	2.944815	0.957524
	Worst	0	0.023715	0.224638	39.36212	1.086747	184.891	0.030204	0	0	0	4.649402	1.023951
	Std	0	0.009821	0.044595	11.27299	0.01329	57.96801	0.009796	0	0	0	0.788509	0.059627
F12	Best	0.00042	5.094209	4.442921	203,911.7	2.172469	572.7398	0.038673	0.006573	2.05E-09	2.52E-08	5.224863	1.372818
	Mean	0.000587	10.6096	8.628854	8,210,509	5.829395	810,249.7	0.097574	0.018775	3.3E-09	1.13E-07	12.35002	1.908009
	Worst	0.000901	18.86702	18.33786	57,132,888	11.33018	4,434,882	0.145051	0.039853	4.68E-09	5.01E-07	19.70848	2.665093
	Std	0.000138	5.243874	4.060513	17,467,648	2.893219	1,503,950	0.036921	0.011515	8.93E-10	1.41E-07	5.340621	0.442747
F13	Best	0.007846	3.953745	43.12097	290,905.2	0.39604	35,635.68	1.319047	0.318217	6.7E-08	0.109128	66.41409	2.197696
	Mean	0.024254	5.014923	63.30228	21,854,085	1.028946	82,687,306	1.852244	0.856902	0.006402	0.323894	80.21083	4.76924
	Worst	0.053684	5.849648	91.49223	75,902,909	1.720694	4.11E + 08	2.405556	1.654764	0.021024	0.882735	111.6855	6.717124
	Std	0.018055	0.575313	16.51973	23,702,134	0.442204	1.73E + 08	0.300261	0.378988	0.008917	0.224292	15.90642	1.558853

Therefore, it is concluded that the STA technique shows the best performance compared to other methods.

3.9.2 Friedman's rank test results

Table 21 shows the statistical results attained by Friedman tests. From the results, we can get the ranks of eleven techniques as follows: STA, INFO, MFO, SSA, MVO, RUN, PSO, DE, GWO, WOA, TSA, and SCA.

To illustrate the ranking of all the compared techniques for each of the ten benchmark functions, a radar chart (Fig. 12) is employed. This outcome provides additional confirmation of our algorithm's effectiveness in uncovering global optima for a variety of problems.

In Fig. 13, the mean ranks derived from the Friedman test for the CEC 2019 benchmark functions using various algorithms are presented. This graph offers a comprehensive comparison of the algorithms' performance on these demanding benchmark functions. Mean ranks provide valuable insights into the relative effectiveness of each algorithm, facilitating a more transparent assessment of their overall performance.

4 STA for engineering design problems

In this section, we employ the proposed STA algorithm to address a series of classical engineering problems characterized by diverse constraints, encompassing both equalities and inequalities. We evaluate the performance of the STA technique in comparison to commonly utilized optimization methods when tackling these engineering design problems. The chosen problems for analysis are widely recognized in the field and encompass the three-bar truss design (BTD), cantilever beam design (CBD), welded beam design (WBD), speed reducer design (SRD), and gear train design (GTD). Metaheuristic algorithms have demonstrated their effectiveness in resolving such constrained engineering challenges. To assess the efficiency of the proposed STA algorithm, we compared its outcomes with those achieved by other metaheuristic algorithms, which include the original TSA, SSA, SCA, MVO, MFO, GWO, INFO, RUN, PSO, DE, and WOA algorithms. For consistency, we maintain the same simulation assumptions as outlined in the preceding section.

4.1 Three-bar truss design problem

The three-bar truss design is a well-known problem in civil engineering, necessitating the adjustment of two parameters to achieve the minimal weight when creating a truss [32]. Figure 14 provides a visual representation of

Table 7 Numerical results of test functions by the STA technique and other recent techniques (Dim = 100)

Function		STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F1	Best	0	4.1E-12	378.3603	865.5214	107.5467	11,857.36	1.73E-14	1.27E-89	3.5E-195	4.22E-54	2161.874	2620.355
	Mean	7.8E-287	9.89E-11	591.4813	7917.665	127.9969	52,881.83	5.04E-14	1.91E-79	2.6E-168	5.99E-53	4155.112	2948.598
	Worst	7.8E-286	5.06E-10	860.3858	16,946.95	158.5009	80,203.25	1.28E-13	1.52E-78	2.6E-167	1.35E-52	6866.153	3363.482
	Std	0	1.59E-10	165.2978	5603.092	17.91651	21,214.56	3.53E-14	4.8E-79	0	5.21E-53	1309.889	243.2402
F2	Best	5.1E-168	1.1E-08	28.51299	1.345816	48,230.13	188.6975	3.06E-09	4.42E-57	3.4E-102	3.67E-26	46.23552	47.03475
	Mean	8E-150	8.22E-08	36.22312	7.734513	3.21E + 26	236.7506	6.99E-09	1.36E-52	1.3E-96	6.23E-26	64.38809	52.8432
	Worst	5.5E-149	1.93E-07	45.19741	28.26774	3.21E + 27	283.4226	1.83E-08	1.33E-51	7.95E-96	1.12E-25	85.95636	64.26155
	Std	1.8E-149	5.17E-08	6.087066	8.238584	1.02E + 27	32.66327	4.47E-09	4.18E-52	2.66E-96	2.54E-26	13.48897	5.590189
F3	Best	0	1608.512	20,124.78	132,024.1	49,764.28	166,062.3	13.83577	710,176.1	2.1E-158	4.06E-50	32,142.77	511,471.7
	Mean	2.7E-287	7075.585	41,778.53	232,124.5	63,084.45	248,569.8	361.7355	912,806.9	5.2E-152	3.25E-49	47,700.97	618,769.8
	Worst	2.7E-286	11,754.09	89,298.81	358,011.9	73,644.13	342,151.5	1618.854	1,282,225	2.8E-151	9.85E-49	65,843	747,777.6
	Std	0	3349.821	22,841.31	70,510.1	7308.01	57,281.1	510.3926	168,393.6	8.9E-152	3.86E-49	11,339.44	72,058.66
F4	Best	2.2E-163	42.93747	19.91026	83.4743	54.53998	87.87274	0.039943	28.97343	1.55E-92	6.71E-28	26.77773	87.18888
	Mean	1.5E-146	51.81269	24.81886	88.50527	58.03784	91.72805	0.381645	76.93045	6.65E-83	2.03E-27	31.00809	89.83904
	Worst	1.5E-145	83.09193	31.20894	93.22804	65.78134	94.63268	1.83412	95.93349	4.21E-82	4.3E-27	38.01327	92.74066
	Std	4.9E-146	11.44159	3.639748	2.793757	3.471043	2.627332	0.534005	24.70817	1.3E-82	1.11E-27	3.483549	1.649486
F5	Best	0.980602	97.74535	29,294.79	38,988,583	2674.524	93,050,917	95.93029	97.51317	94.55342	91.64101	481,832	3,773,920
	Mean	6.332139	98.3651	59,116.76	1.19E + 08	6981.168	1.67E + 08	97.81836	98.09964	96.68201	93.78239	946,600.2	6,506,272
	Worst	15.74872	98.62866	113,857.6	1.81E + 08	33,372.45	2.39E + 08	98.51145	98.27593	98.25104	95.42777	1,681,374	11,653,848
	Std	5.64575	0.313931	28,592.57	47,008,190	9349.51	40,860,489	0.910646	0.232995	1.265715	1.157989	417,217.5	2,220,964
F6	Best	0.034199	12.36507	458.3328	3382.668	97.81795	34,136.61	6.657808	1.727985	5.39E-07	1.014726	3394.583	2313.334
	Mean	0.047486	14.14842	734.3811	9346.223	121.936	60,976.1	8.138721	2.985413	0.000158	1.55265	4440.483	2805.015
	Worst	0.078961	16.9627	1179.041	14,853.74	136.651	106,398	9.90864	5.384838	0.001494	2.305538	5885.616	3067.965
	Std	0.012886	1.324649	205.5416	4913.27	12.46786	21,128.22	0.952747	1.02795	0.00047	0.426929	766.788	277.8063
F7	Best	8.3E-07	0.016629	1.087689	47.97846	0.306437	57.07258	0.003488	0.000103	1.58E-05	9.96E-05	2.58486	7.063484
	Mean	2.17E-05	0.043235	1.845356	170.1335	0.524315	236.0788	0.004872	0.002072	0.00023	0.000957	5.050338	9.035567
	Worst	6.22E-05	0.075556	2.908462	264.2855	0.668304	423.0744	0.00719	0.006339	0.000587	0.001791	10.83312	11.63818
	Std	1.88E-05	0.018899	0.575068	70.18115	0.128859	133.4702	0.001002	0.002278	0.000198	0.000554	2.269359	1.733864
F8	Best	− 41,865.5	− 14,540.5	− 25,242.7	− 8919.39	− 25,411.4	− 25,941.3	− 19,934.6	− 41,895.4	− 28,015.5	− 28,036.7	− 21,731.5	− 18,529.9
	Mean	− 41,774.2	− 13,229.4	− 22,321.2	− 7451.65	− 23,576.4	− 23,186.5	− 15,738	− 36,736.6	− 24,692.6	− 26,211.7	− 18,082.3	− 16,666.4
	Worst	− 41,543.9	− 11,903.3	− 18,851.2	− 6126.82	− 22,336.4	− 18,641.6	− 6659.6	− 29,091.2	− 20,805.6	− 24,028.5	− 14,534.4	− 14,534.4
	Std	107.3667	868.4634	2346.993	947.7936	1137.049	2121.292	3591.254	5334.586	2286.131	1662.977	2128.227	1491.002
F9	Best	0	773.7447	125.2122	152.5709	566.3136	664.5154	6.14E-11	0	0	0	256.4373	738.8997
	Mean	0	956.2932	176.3458	263.5528	643.0111	799.0757	9.651146	0	0	0	346.8623	811.1128
	Worst	0	1204.263	272.667	476.2615	776.2912	892.8116	18.24781	0	0	0	443.1879	872.0398
	Std	0	139.0176	46.48856	115.0505	80.23399	70.41346	6.234152	0	0	0	55.03576	38.4504

Table 7 (continued)

Function		STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F10	Best	8.88E-16	2.56E-07	6.823348	1.715173	3.956072	19.46019	1.7E-08	8.88E-16	8.88E-16	8.88E-16	9.929306	8.105028
	Mean	8.88E-16	2.54E-06	8.470158	17.99463	10.77182	19.84911	3.1E-08	4.44E-15	8.88E-16	8.88E-16	11.23433	8.529307
	Worst	8.88E-16	6.97E-06	11.95637	20.64303	20.26549	19.95754	4.59E-08	7.99E-15	8.88E-16	8.88E-16	13.06912	8.822892
	Std	0	2.07E-06	1.581998	6.140519	8.039878	0.17521	1.02E-08	3.35E-15	0	0	1.04149	0.288819
F11	Best	0	4.57E-12	2.94161	9.838384	1.947822	363.1009	1.49E-14	0	0	0	28.30909	21.34351
	Mean	0	0.003107	6.621054	100.8828	2.066827	505.6021	3.87E-14	0	0	0	39.41129	28.2486
	Worst	0	0.031074	10.37547	203.679	2.476388	708.3487	1.31E-13	0	0	0	52.26416	32.03504
	Std	0	0.009826	2.299602	72.36172	0.156005	120.593	3.41E-14	0	0	0	9.344024	3.263482
F12	Best	0.000424	5.415622	14.14765	1.01E + 08	11.53487	49,188,224	0.111616	0.016963	3.28E-08	0.002051	39.71103	4,292,849
	Mean	0.000675	12.28207	26.21394	3.04E + 08	16.7856	2.82E + 08	0.243176	0.029446	7.67E-08	0.006856	1122.436	9,405,800
	Worst	0.00085	20.15531	49.16219	5.96E + 08	27.1092	6.9E + 08	0.424833	0.040757	2.24E-07	0.017215	6683.961	20,003,600
	Std	0.000132	5.129635	11.13134	1.45E + 08	4.570245	2.21E + 08	0.075968	0.007438	5.82E-08	0.004643	2224.137	4,525,187
F13	Best	0.017214	10.36722	182.811	1.13E + 08	107.0288	2.13E + 08	5.587851	0.609042	0.00025	2.946222	43,123.9	13,059,823
	Mean	0.039222	12.43319	493.9032	5.39E + 08	154.832	6.67E + 08	6.493904	1.740396	0.025125	5.598381	188,058.2	22,030,774
	Worst	0.060058	15.45363	1683.419	8.77E + 08	192.7361	1.11E + 09	7.155638	2.707473	0.068181	9.897633	560,427.6	31,427,543
	Std	0.01315	1.386865	510.5826	2.75E + 08	23.44756	3.37E + 08	0.479376	0.655007	0.022548	2.371115	161,403.6	5,828,554

this problem. The mathematical model for this design is expressed as follows.

$$\text{Minimize } f(A1, A2) = l \times \left(2\sqrt{2}x_1 + x_2\right)$$

$$\text{Subject to } G_1 = \frac{\sqrt{2}x_1 + x_2}{\sqrt{2}x_1^2 + 2x_1x_2}P - \sigma \leq 0,$$

$$G_2 = \frac{x_2}{\sqrt{2}x_1^2 + 2x_1x_2}P - \sigma \leq 0$$

$$G_3 = \frac{1}{\sqrt{2}x_2 + x_1}P - \sigma \leq 0$$

In these equations, $l = 100\,cm; P = \frac{2kN}{\mathrm{cm}^2}; \sigma = \frac{2kN}{\mathrm{cm}^2}$.

Variable range: $0 \leq x_1, x_2 \leq 1.00$,

This optimization problem holds significant importance in civil engineering applications as it aims to attain an efficient and lightweight design for truss structures. Table 22 shows the performance of several algorithms, such as TSA, SSA, SCA, MVO, MFO, GWO, WOA, INFO, RUN, PSO, DE, and the proposed STA algorithm as well as the algorithms employed to solve three-bar truss design issue including ray and sain [33], artificial atom algorithm (AAA) [34], mine blast algorithm (MBA) [35], differential evolution with dynamic stochastic selection (DEDS) [36], grasshopper optimization algorithm (GOA) [37], hybridizing particle swarm optimization with differential evolution (PSO-DE) [38], and cuckoo search algorithm (CS) [39]. The results reveal that the STA algorithm outperforms the other algorithms. It achieves a minimum weight of 263.895, with optimal values for × 1 and × 2 being 0.788675 and 0.408248, respectively. Figure 15 displays the convergence curves and the boxplot for these algorithms while solving the problem. Furthermore, Table 23 provides a comparative analysis of the statistical results, with the mean and standard deviation values obtained by the proposed STA algorithm being superior to all other counterparts.

4.2 Cantilever beam design problem

The objective of the cantilever beam design problem is to reduce the weight of a cantilever with five hollow blocks, as depicted in Fig. 16 [40]. This involves five variables, and the mathematical representation of the structural optimization problem is as follows:

$$\text{Consider: } \vec{x} = [x_1x_2x_3x_4x_5]$$

$$\text{Minimize } f(X) = 0.0624(x_1 + x_2 + x_3 + x_4 + x_5)$$

$$\text{Subject to } G(X) = \frac{61}{x_1^3} + \frac{37}{x_2^3} + \frac{19}{x_3^3} + \frac{7}{x_4^3} + \frac{1}{x_5^3} - 1 \leq 0,$$

$$\text{Variable range:} 0.01 \leq x_i \leq 1.00; i \in 1, \ldots, 5.$$

Table 24 tabulates the optimal solutions to this problem attained by the STA optimizer and the results are compared with these of recent techniques and the algorithms used to solve cantilever beam design problem

Table 8 Numerical results of test functions by the STA technique and other recent techniques (Dim = 500)

Function		STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F1	Best	0	0.001893	62,067.97	106,814.6	69,212.2	1,101,214	0.000359	7.76E-85	3.4E-185	1.12E-53	131,315.6	1,464,623
	Mean	6.2E-293	0.012947	70,965.19	164,070.5	88,323.78	1,162,497	0.000548	2.94E-76	3.8E-164	2.05E-52	159,498.3	1,506,774
	Worst	6.2E-292	0.029717	77,541.7	228,785	100,655.4	1,222,267	0.00115	2.68E-75	2.9E-163	6.18E-52	173,739.4	1,555,415
	Std	0	0.009072	5173.569	35,208.1	9127.749	44,543.91	0.000237	8.42E-76	0	2.16E-52	13,386.15	26,595.93
F2	Best	8.3E-165	0.004845	431.8474	23.0235	1.8E + 154	1.64E + 73	0.004109	2.76E-55	7.4E-99	1.4E-25	653.4217	1.3E + 213
	Mean	1.2E-153	0.006862	462.2622	66.5909	1.9E + 218	2.2E + 109	0.005531	1.19E-50	5.55E-89	2.82E-25	972.8254	3E + 223
	Worst	1.2E-152	0.014331	505.5605	138.8087	1.9E + 219	2.2E + 110	0.00757	9.49E-50	5.49E-88	4.33E-25	1169.916	3E + 224
	Std	3.7E-153	0.002746	21.20328	37.54569	65,535	7E + 109	0.001046	2.94E-50	1.73E-88	1.06E-25	189.8349	65,535
F3	Best	0	982,117.5	793,322.8	6,570,642	1,598,513	4,200,822	235,225.7	11,310,381	1.5E-147	1.29E-47	608,338.4	13,133,362
	Mean	2.6E-302	1,367,741	1,115,118	7,144,948	1,820,255	5,086,544	334,609.9	19,559,475	4.6E-136	4.71E-47	1,066,890	15,278,384
	Worst	1.3E-301	1,693,425	1,357,365	8,053,920	1,988,160	6,906,083	443,497.7	31,840,128	3E-135	1.3E-46	1,745,694	19,277,264
	Std	0	310,292.6	206,711.4	673,089.4	157,934.3	1,257,236	99,718.99	7,690,346	9.6E-136	3.6E-47	353,229.7	1,821,670
F4	Best	6.8E-175	99.04009	32.1551	98.81637	91.87189	98.62133	51.75566	41.75266	2.19E-84	1.91E-28	41.0117	98.80389
	Mean	2.8E-140	99.20716	36.09473	99.11544	94.03506	98.77144	62.76924	87.66102	1.49E-78	1.18E-27	51.75798	99.21992
	Worst	2.8E-139	99.41109	45.29265	99.41946	95.58507	98.92027	67.84426	98.97866	1.12E-77	2.93E-27	60.56493	99.46512
	Std	8.7E-140	0.123676	4.162495	0.190769	1.17558	0.115321	4.456214	20.38406	3.49E-78	7.57E-28	6.526692	0.202991
F5	Best	3.609439	1380.159	16,697,856	1.41E + 09	91,774,361	4.61E + 09	497.4158	495.3234	492.6181	496.5455	74,741,695	6.59E + 09
	Mean	62.31615	43,487.94	20,515,939	1.74E + 09	1.14E + 08	4.93E + 09	497.7697	496.0706	494.2847	496.7852	1.05E + 08	7.04E + 09
	Worst	109.4461	172,168.2	26,196,244	2.21E + 09	1.49E + 08	5.39E + 09	497.9476	496.7729	495.2495	497.0414	1.87E + 08	7.33E + 09
	Std	42.8145	53,479.37	2,642,841	2.58E + 08	20,678,691	2.33E + 08	0.161382	0.518417	0.95265	0.177492	32,812,785	2.58E + 08
F6	Best	0.39516	97.73253	66,769.06	73,047.18	78,253.23	1,099,287	86.35288	17.89194	1.083312	64.56382	137,787.5	1,473,120
	Mean	0.511359	101.082	72,212.41	182,361.7	88,432.07	1,151,232	88.78923	23.78796	1.478187	69.18278	159,874.4	1,501,572
	Worst	0.679021	105.8421	76,918.46	338,000.4	98,992.73	1,190,893	90.52986	29.57335	1.751566	74.76969	172,833.2	1,548,503
	Std	0.097551	2.843378	3445.456	88,436.46	6057.424	31,783.97	1.352624	4.145	0.205294	3.277371	13,099.22	26,284.19
F7	Best	7.84E-07	0.762517	136.5866	12,228.86	619.6518	31,918.48	0.023291	3.12E-05	2.93E-05	0.000472	515.074	52,795.27
	Mean	2.61E-05	1.528576	163.513	15,570.37	727.3755	36,364.8	0.035312	0.003164	0.000219	0.001261	897.7774	57,172.89
	Worst	8.44E-05	3.089053	203.6756	22,352.78	869.8485	41,340.88	0.054219	0.013534	0.000567	0.002476	1631.962	60,363.92
	Std	2.88E-05	0.778756	21.09103	3538.065	85.65161	3051.979	0.009082	0.004289	0.000168	0.000732	308.0781	2510.143
F8	Best	− 209,176	− 35,882.3	− 73,894.7	− 16,861	− 80,519.2	− 72,783.2	− 71,967	− 209,455	− 123,368	− 102,763	− 58,698.2	− 58,698.2
	Mean	− 208,141	− 33,300.4	− 64,945	− 15,768.2	− 78,129.9	− 62,313.8	− 60,298.7	− 193,972	− 97,119.3	− 98,199.3	− 50,711.4	− 52,918.2
	Worst	− 205,290	− 29,201.6	− 58,729.3	− 14,251.3	− 75,079.6	− 55,272.9	− 51,719	− 144,600	− 74,593.7	− 87,834.6	− 46,001.8	− 50,419.5
	Std	1293.724	2423.981	4717.24	929.4627	1826.625	5465.831	6279.925	26,028.9	17,220.69	4238.092	3493.398	3145.718

Table 8 (continued)

Function		STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F9	Best	0	5228.249	2723.724	379.8043	5911.853	6521.588	35.18143	0	0	0	3419.589	8504.725
	Mean	0	5945.568	2915.717	1133.231	6321.152	6897.345	50.3377	0	0	0	3775.655	8652.841
	Worst	0	6630.217	3093.985	2161.389	6720.62	7225.044	82.37053	0	0	0	4050.472	8750.344
	Std	0	537.8106	140.248	541.1296	213.462	180.8596	13.58614	0	0	0	195.9292	72.78386
F10	Best	8.88E-16	0.003274	12.754	14.06618	20.78006	20.01038	0.000672	8.88E-16	8.88E-16	8.88E-16	15.27714	21.03636
	Mean	8.88E-16	0.005354	13.35621	19.46291	20.82752	20.25615	0.000922	4.09E-15	8.88E-16	8.88E-16	15.88607	21.0774
	Worst	8.88E-16	0.008477	13.9354	20.83839	20.8994	20.38568	0.001168	7.99E-15	8.88E-16	8.88E-16	16.39356	21.10121
	Std	0	0.00169	0.329336	2.78739	0.032963	0.138704	0.000186	2.02E-15	0	0	0.322423	0.020788
F11	Best	0	0.000557	579.5803	1189.001	738.9494	10,055.73	3.87E-05	0	0	0	1239.765	13,169.59
	Mean	0	0.001608	637.3652	1881.146	801.9722	10,331.76	0.008048	0	0	0	1453.111	13,558.8
	Worst	0	0.003827	722.2998	3048.317	882.829	10,699.85	0.080008	0	0	0	1781.849	13,854.98
	Std	0	0.000978	40.84853	604.472	45.31254	233.1235	0.025284	0	0	0	155.7532	248.8978
F12	Best	0.000377	173,650.5	38,856.61	4.78E + 09	44,293,353	9.94E + 09	0.653806	0.017164	0.001209	0.254133	17,481,355	1.64E + 10
	Mean	0.000706	600,131.9	172,645.9	6.19E + 09	86,592,912	1.19E + 10	0.693955	0.052096	0.001481	0.299525	40,707,475	1.74E + 10
	Worst	0.000905	2,906,619	301,758.8	7.07E + 09	1.37E + 08	1.32E + 10	0.759184	0.082509	0.001726	0.363503	1.09E + 08	1.81E + 10
	Std	0.000189	823,882.4	98,105.54	7.77E + 08	27,567,071	8.58E + 08	0.036453	0.020677	0.000162	0.028821	27,835,510	5.9E + 08
F13	Best	0.163863	168,109.9	8,660,548	5.83E + 09	1.76E + 08	2.06E + 10	47.09021	6.00445	1.251251	49.57657	1.45E + 08	3.14E + 10
	Mean	0.230713	314,775.2	14,536,381	9.31E + 09	2.69E + 08	2.2E + 10	49.19635	13.06554	2.889947	49.74887	2.15E + 08	3.24E + 10
	Worst	0.289573	520,668.8	21,701,001	1.13E + 10	3.76E + 08	2.33E + 10	50.50283	22.98249	3.922078	49.78513	3.2E + 08	3.39E + 10
	Std	0.037716	130,224.5	4,132,938	1.88E + 09	59,799,447	9.37E + 08	1.174419	4.428498	0.836929	0.06128	60,140,174	7.06E + 08

Table 9 Numerical results of composite test functions by the STA technique and other recent techniques

Function		STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F14	Best	0.998004	0.998004	0.998004	0.998025	0.998004	0.998004	0.998004	0.998004	0.998004	0.998004	0.998004	0.998004
	Mean	1.887908	9.467442	1.196414	2.188736	1.689498	1.887908	4.721197	1.891047	3.844082	2.76463	2.384526	1.887908
	Worst	5.928845	15.50382	2.982105	2.982105	5.928845	5.928845	10.76318	2.982105	10.76318	10.76318	5.928845	5.928845
	Std	1.642059	5.128187	0.627428	1.024235	1.6148	1.642059	4.258338	0.986546	3.74834	3.287764	1.485597	1.642059
F15	Best	0.000307	0.000308	0.000502	0.000484	0.000526	0.000566	0.000309	0.000317	0.000307	0.000307	0.000338	0.000857
	Mean	0.000309	0.008453	0.00283	0.001099	0.000669	0.001656	0.008364	0.000839	0.000765	0.000674	0.000542	0.000959
	Worst	0.000317	0.020363	0.020363	0.001559	0.00074	0.008334	0.020363	0.002252	0.001223	0.001223	0.000746	0.001134
	Std	3.13E-06	0.010254	0.006165	0.000445	8.24E-05	0.002367	0.010328	0.000607	0.000483	0.000473	0.000114	9.82E-05
F16	Best	− 1.03163	− 1.03163	− 1.03163	− 1.03163	− 1.03163	− 1.03163	− 1.03163	− 1.03163	− 1.03163	− 1.03163	− 1.03163	− 1.03163
	Mean	− 1.03163	− 1.03163	− 1.03163	− 1.0316	− 1.03163	− 1.03162	− 1.03162	− 1.03163	− 1.03163	− 1.03163	− 1.03163	− 1.03163
	Worst	− 1.03163	− 1.03163	− 1.03163	− 1.03155	− 1.03163	− 1.03159	− 1.03159	− 1.03163	− 1.03163	− 1.03163	− 1.03163	− 1.03163
	Std	1.96E-16	5.4E-07	1.51E-14	2.75E-05	3.27E-07	1.23E-05	1.23E-05	5.77E-10	1.72E-13	5.35E-15	5.6E-15	5.35E-15
F17	Best	0.397887	0.397888	0.397887	0.398085	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887	0.397887
	Mean	0.397887	0.397912	0.397887	0.399802	0.397888	0.397887	0.397896	0.397893	0.397887	0.39809	0.39809	0.397888
	Worst	0.397887	0.397969	0.397887	0.40375	0.397889	0.397888	0.397955	0.397916	0.397887	0.398959	0.398959	0.397895
	Std	0	2.5E-05	2.68E-14	0.001765	3.74E-07	2.83E-07	2.1E-05	1.08E-05	3.04E-11	0.000429	0.000429	2.43E-06
F18	Best	3	3.000008	3	3	3	3	3	3	3	3	3	3
	Mean	3	8.400024	3	3.000034	3.000004	3	3.000011	3.000013	3	3	3	3
	Worst	3	30.00006	3	3.000095	3.000016	3.000001	3.000024	3.000041	3.000001	3.000001	3.000001	3.000001
	Std	1.97E-15	11.38421	2.55E-13	3.36E-05	5.05E-06	3.25E-07	8.34E-06	1.52E-05	2.43E-07	2.43E-07	2.43E-07	2.43E-07
F19	Best	− 3.86278	− 3.86277	− 3.86278	− 3.86184	− 3.86278	− 3.86278	− 3.86278	− 3.86271	− 3.86278	− 3.86278	− 3.86278	− 3.86278
	Mean	− 3.86278	− 3.86195	− 3.86278	− 3.85578	− 3.86278	− 3.86278	− 3.86194	− 3.85756	− 3.86278	− 3.86199	− 3.86278	− 3.86278
	Worst	− 3.86278	− 3.8549	− 3.86278	− 3.85088	− 3.86278	− 3.86277	− 3.85492	− 3.82596	− 3.86278	− 3.85492	− 3.86277	− 3.86277
	Std	6.55E-10	0.002477	7.43E-14	0.004038	7.37E-07	2.75E-06	0.00247	0.011236	6.13E-09	0.002485	3.66E-06	3.66E-06
F20	Best	− 3.322	− 3.32115	− 3.322	− 3.21771	− 3.32199	− 3.322	− 3.32199	− 3.32113	− 3.322	− 3.322	− 3.322	− 3.322
	Mean	− 3.28611	− 3.26789	− 3.22065	− 2.86738	− 3.26193	− 3.25066	− 3.25552	− 3.2376	− 3.23877	− 3.26255	− 3.27444	− 3.26892
	Worst	− 3.20114	− 3.15398	− 3.18014	− 2.24178	− 3.20085	− 3.2031	− 3.08056	− 3.04064	− 3.2031	− 3.2031	− 3.2031	− 3.20114
	Std	0.057778	0.069004	0.053797	0.304519	0.063315	0.061396	0.092273	0.112977	0.057431	0.062662	0.061396	0.059554
F21	Best	− 10.1532	− 10.1461	− 10.1532	− 4.85594	− 10.1531	− 10.1532	− 10.152	− 10.153	− 10.1532	− 10.1532	− 10.1532	− 10.1532
	Mean	− 9.6434	− 5.10638	− 8.39112	− 3.1047	− 8.63267	− 7.13429	− 8.89423	− 7.3576	− 10.1532	− 9.5536	− 8.90092	− 9.77729
	Worst	− 5.0552	− 2.61976	− 2.68286	− 0.87829	− 5.05513	− 2.68286	− 2.63042	− 2.62996	− 10.1532	− 5.0552	− 2.68286	− 7.31463
	Std	1.61213	3.477945	2.911156	1.750653	2.447805	3.299494	2.713752	3.02625	9.86E − 10	1.605544	2.700844	0.90985
F22	Best	− 10.4029	− 10.2978	− 10.4029	− 4.96823	− 10.4028	− 10.4029	− 10.4023	− 10.3997	− 10.4029	− 10.4029	− 10.4029	− 10.4029
	Mean	− 10.4029	− 7.33299	− 8.10903	− 2.71188	− 8.04437	− 9.73501	− 10.4016	− 7.52469	− 9.25733	− 7.67612	− 8.344	− 10.4018
	Worst	− 10.4029	− 1.82993	− 2.75193	− 0.90712	− 2.76589	− 3.7243	− 10.4002	− 1.83567	− 1.83567	− 2.7659	− 2.7659	− 10.3965
	Std	5.7E − 08	3.771518	3.693541	1.880192	3.117236	2.111947	0.000675	3.332978	2.760538	3.57696	3.374895	0.002073

Table 9 (continued)

Function		STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F23	Best	− 10.5364	− 10.3575	− 10.5364	− 9.39089	− 10.5363	− 10.5364	− 10.536	− 10.5359	− 10.5364	− 10.5364	− 10.5364	− 10.5364
	Mean	− 10.0003	− 4.33786	− 9.99562	− 5.10027	− 8.21714	− 8.54781	− 10.5348	− 8.13812	− 9.19621	− 7.71521	− 7.70842	− 8.51933
	Worst	− 5.17565	− 1.85281	− 5.12848	− 3.07402	− 2.8066	− 2.87114	− 10.5331	− 2.8065	− 3.83543	− 2.42734	− 2.42734	− 2.42734
	Std	1.695222	3.257715	1.710137	1.609525	3.733619	3.241771	0.000965	3.162138	2.825383	3.664768	3.729461	3.311793

including generalized convex approximation (GCA) [41], ant lion optimizer (ALO) [14], method of moving asymptotes (MMA) [41], symbiotic organisms search (SOS) [42], and improved slime mold algorithm (ISMA) [43]. The results demonstrate that the STA technique efficiently solves the problem and the design with the minimum weight $\vec{x} = [6.014644 \quad 5.310063 \quad 4.494105 \quad 3.501842 \quad 2.153006]$.

Figure 17 illustrates the convergence curves and box plots for all algorithms when dealing with the Cantilever Beam Design problem. Table 25 presents the statistical information, encompassing best, mean, median, worst, and STD. Consequently, it can be inferred that the STA optimizer exhibits greater consistency in addressing this problem.

4.3 Welded beam design

The aim of this engineering problem is to minimize the production cost of a welded beam [44]. It entails the consideration of several optimization constraints, including shear stress (τ), buckling load on the bar (Pc), end deflection of the beam (δ), and bending stress in the beam (θ). The problem encompasses four variables: the thickness of the weld (h), the length of the attached part to the bar (l), the height of the bar (t), and the thickness of the bar (b) (as depicted in Fig. 18). Mathematically, the problem can be expressed as follows.

$$\begin{aligned} &\text{Consider}: \ \vec{x} = [x_1 x_2 x_3 x_4] = [hltb], \\ &\text{Minimize} f(\vec{x}) = 1.1047 x_1^2 x_2 + 0.04811 x_3 x_4 (14.0 + x_2), \\ &\text{Subject to } g_1(\vec{x}) = \tau(\vec{x}) - \tau_{\max} \le 0, \end{aligned}$$

$$\begin{aligned} &g_2(\vec{x}) = \sigma(\vec{x}) - \sigma_{\max} \le 0 \\ &g_3(\vec{x}) = \delta(\vec{x}) - \delta_{\max} \le 0, \\ &g_4(\vec{x}) = x_1 - x_4 \le 0, \\ &g_5(\vec{x}) = p - p_c(\vec{x}) \le 0, \\ &g_6(\vec{x}) = 0.125 - x_1 \le 0, \\ &g_7(\vec{x}) = 1.10471 x_1^2 x_2 - 0.0481 x_3 x_4 (14.0 + x_2) - 5.0 \le 0 \end{aligned}$$

Variable range: $0.1 \le x_1 \le 2.00$,

$0.1 \le x_2 \le 10.0$,

$0.1 \le x_3 \le 10.0$,

$0.1 \le x_4 \le 2.00$,

Table 26 shows the results achieved from the analysis, representing that the STA technique outperforms other techniques and the algorithms used to solve welded beam design problem including such as GOA [37], *ray optimization* (RO) [45], effective co-evolutionary differential evolution (ECDE) [46], genetic algorithm (GA) [47],

Table 10 Statistical results of the Wilcoxon rank-sum test (Dim = 50)

STA vs Fun	TSA		SSA		SCA		MVO		MFO	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
F1	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+
F2	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F3	8.74E-05	+	8.74E-05	+	8.74E-05	+	8.74E-05	+	8.74E-05	+
F4	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F5	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F6	1.83E-04	+	1.73E-02	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F7	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F8	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F9	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+
F10	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+
F11	2.31E-04	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+
F12	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F13	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
WRST (+ / = / −)	13/0/0		13/0/0		13/0/0		13/0/0		13/0/0	

STA vs Fun	GWO		WOA		RUN		INFO		PSO		DE	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
F1	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+
F2	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F3	8.74E-05	+	8.74E-05	+	8.74E-05	+	8.74E-05	+	8.74E-05	+	8.74E-05	+
F4	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F5	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F6	1.83E-04	+	1.83E-04	+	1.83E-04	-	1.83E-04	-	1.83E-04	+	1.83E-04	+
F7	1.83E-04	+	1.83E-04	+	7.69E-04	+	2.46E-04	+	1.83E-04	+	1.83E-04	+
F8	1.83E-04	+	3.45E-01	=	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F9	6.39E-05	+	NaN	=	NaN	=	NaN	=	6.39E-05	+	6.39E-05	+
F10	6.39E-05	+	1.43E-02	+	NaN	=	NaN	=	6.39E-05	+	6.39E-05	+
F11	1.68E-01	=	NaN	=	NaN	=	NaN	=	6.39E-05	+	6.39E-05	+
F12	1.83E-04	+	1.83E-04	+	1.83E-04	-	1.83E-04	-	1.83E-04	+	1.83E-04	+
F13	1.83E-04	+	1.83E-04	+	1.13E-02	-	1.83E-04	+	1.83E-04	+	1.83E-04	+
WRST (+ / = / −)	12/1/0		10/3/0		7/3/3		8/3/2		13/0/0		13/0/0	

Table 11 Statistical results of the Wilcoxon rank-sum test (Dim = 100)

STA vs Fun	TSA		SSA		SCA		MVO		MFO	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
F1	1.32E-04	+	1.32E-04	+	1.32E-04	+	1.32E-04	+	1.32E-04	+
F2	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F3	8.74E-05	+	8.74E-05	+	8.74E-05	+	8.74E-05	+	8.74E-05	+
F4	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F5	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F6	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F7	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F8	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F9	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+
F10	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+
F11	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+
F12	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F13	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
WRST (+ / = / −)	13/0/0		13/0/0		13/0/0		13/0/0		13/0/0	

STA vs Fun	GWO		WOA		RUN		INFO		PSO		DE	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
F1	1.32E-04	+	1.32E-04	+	1.32E-04	+	1.32E-04	+	1.32E-04	+	1.32E-04	+
F2	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F3	8.74E-05	+	8.74E-05	+	8.74E-05	+	8.74E-05	+	8.74E-05	+	8.74E-05	+
F4	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F5	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F6	1.83E-04	+	1.83E-04	+	1.83E-04	–	1.83E-04	+	1.83E-04	+	1.83E-04	+
F7	1.83E-04	+	1.83E-04	+	1.01E-03	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F8	1.83E-04	+	2.83E-03	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.73E-04	+
F9	6.39E-05	+		=	NaN	=	NaN	=	6.39E-05	+	6.39E-05	+
F10	6.39E-05	+	5.66E-03	+	NaN	=	NaN	=	6.39E-05	+	6.39E-05	+
F11	6.39E-05	+		=	NaN	=	NaN	=	6.39E-05	+	6.39E-05	+
F12	1.83E-04	+	1.83E-04	+	1.83E-04	–	1.83E-04	+	1.83E-04	+	1.83E-04	+
F13	1.83E-04	+	1.83E-04	+	6.40E-02	=	1.83E-04	+	1.83E-04	+	1.83E-04	+
WRST (+ / = / −)	13/0/0		11/2/0		7/4/2		10/3/0		13/0/0		13/0/0	

Table 12 Statistical results of the Wilcoxon rank-sum test (Dim = 500)

STA vs Fun	TSA		SSA		SCA		MVO		MFO	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
F1	1.63E-04	+	1.63E-04	+	1.63E-04	+	1.63E-04	+	1.63E-04	+
F2	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F3	1.03E-04	+	1.03E-04	+	1.03E-04	+	1.03E-04	+	1.03E-04	+
F4	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F5	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F6	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F7	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F8	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F9	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+
F10	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+
F11	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+
F12	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F13	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
WRST (+ / = / −)	13/0/0		13/0/0		13/0/0		13/0/0		13/0/0	

STA vs Fun	GWO		WOA		RUN		INFO		PSO		DE	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
F1	1.63E-04	+	1.63E-04	+	1.63E-04	+	1.63E-04	+	1.63E-04	+	1.63E-04	+
F2	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F3	1.03E-04	+	1.03E-04	+	1.10E-04	+	1.10E-04	+	1.10E-04	+	1.10E-04	+
F4	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F5	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F6	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F7	1.83E-04	+	1.31E-03	+	1.31E-03	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F8	1.83E-04	+	9.10E-01	=	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.78E-04	+
F9	6.39E-05	+		=		=		=	6.39E-05	+	6.39E-05	+
F10	6.39E-05	+	5.31E-04	+		=		=	6.39E-05	+	6.39E-05	+
F11	6.39E-05	+		=		=		=	6.39E-05	+	6.39E-05	+
F12	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F13	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
WRST (+ / = / −)	13/0/0		10/3/0		10/3/0		10/3/0		13/0/0		13/0/0	

Table 13 Statistical results of the Wilcoxon rank-sum test

STA vs Fun	TSA		SSA		SCA		MVO		MFO	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
F14	7.48E-04	+	5.31E-01	=	1.70E-02	+	8.14E-02	=	1.17E-01	=
F15	2.20E-03	+	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	+
F16	1.29E-04	+	1.28E-04	+	1.29E-04	+	1.29E-04	−	2.26E-02	−
F17	6.39E-05	+	5.89E-05	+	6.39E-05	+	6.39E-05	+	3.68E-01	=
F18	1.71E-04	+	1.71E-04	+	1.71E-04	+	1.71E-04	+	9.38E-01	=
F19	1.63E-04	+	3.29E-02	+	1.63E-04	+	1.63E-04	−	1.60E-03	−
F20	2.57E-02	+	3.76E-02	+	4.40E-04	+	2.11E-02	+	5.64E-01	=
F21	4.40E-04	+	7.91E-01	=	1.83E-04	+	2.20E-03	+	8.49E-01	=
F22	1.83E-04	+	3.45E-01	=	1.83E-04	+	1.83E-04	−	2.38E-02	−
F23	3.30E-04	+	7.34E-01	=	2.46E-04	+	1.31E-03	-	1.35E-01	=
WRST (+ / = / −)	10/0/0		6/4/0		10/0/0		5/1/4		1/6/3	

STA vs Fun	GWO		WOA		RUN		INFO		PSO		DE	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
F14	5.70E-03	+	3.73E-02	+	2.76E-02	+	3.73E-01	=	4.46E-01	=	1.17E-01	=
F15	1.31E-03	+	2.46E-04	+	9.70E-01	=	4.73E-01	=	1.83E-04	+	1.83E-04	+
F16	1.29E-04	+	1.29E-04	+	1.29E-04	+	1.47E-03	+	1.14E-02	+	1.47E-03	+
F17	6.39E-05	+	6.39E-05	+	7.51E-04	+	1.68E-01	=	1.68E-01	=	1.68E-01	=
F18	1.71E-04	+	1.71E-04	+	1.64E-02	+	1.06E-02	–	8.77E-01	=	1.12E-02	−
F19	1.63E-04	+	1.63E-04	+	2.04E-03	+	1.29E-01	=	3.44E-02	–	2.01E-02	−
F20	3.12E-02	+	1.73E-02	+	7.34E-01	=	2.66E-01	=	9.82E-02	=	8.50E-01	=
F21	2.20E-03	+	7.69E-04	+	2.12E-01	=	3.22E-01	=	1.94E-02	−	6.77E-01	=
F22	1.83E-04	+	1.83E-04	+	9.70E-01	=	4.71E-01	=	1.39E-01	=	2.41E-01	=
F23	2.83E-03	+	1.01E-03	+	4.27E-01	=	9.70E-01	=	4.25E-01	=	3.84E-01	=
WRST (+ / = / −)	10/0/0		10/0/0		5/5/0		1/8/1		2/6/2		2/6/2	

harmony search (HS) [48], an effective co-evolutionary particle swarm optimization (CPSO) [49], and marine predators algorithm (MPA) [21] by attaining a design with the minimum cost value; it becomes evident that when the parameters are set to $h = 0.20572964$, $l = 3.47048867$, $t = 9.03662391$, and $b = 0.20572964$, the manufacturing cost of the welded beam design problem can be reduced to 1.72485230859736. Thus, the proposed STA algorithm exhibits promising capabilities in addressing the welded beam design problem.

Figure 19 displays the convergence patterns and box plots of all methods applied to the welded beam design problem. In Table 27, you can find statistical data, including best, mean, median, worst, and STD. Thus, it can be deduced that the STA technique exhibits superior reliability for this problem, and the results underscore the STA algorithm's outperformance when compared to other algorithms including WOA, SSA, SCA, MFO, MVO, GWO, TSA, INFO, RUN, PSO, DE, Rao_1 [50], Rao_2 [50], Rao_3 [50], FISA [50], and MPA [21].

Table 14 Friedman test for the twelve techniques (Dim = 50)

Function	STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F1	1	6	7	10.8	9	12	5	3	2	4	10.2	8
F2	1	6	9.4	7.3	11.1	11.3	5	3	2	4	10.2	7.7
F3	1	5	7.1	9.2	6.4	9.8	4	12	2	3	7.5	11
F4	1	5.2	6.9	10.6	6.4	11.7	4	10.1	2	3	7.9	9.2
F5	1	5.6	7.4	12	7.8	11	4	5.3	2.9	2.2	9.6	9.2
F6	3.2	8.4	3.8	10.9	8.6	11.8	6.4	5	1	2	10.3	6.6
F7	1	6	9.5	11	7	11.8	4.4	4	2.3	3.3	9.7	8
F8	1.4	10.5	7	12	5.3	3.7	9.3	1.9	5.5	3.5	8.6	9.3
F9	2.5	11.7	6.9	7.2	9.5	11.1	5	2.5	2.5	2.5	6.9	9.7
F10	2.25	6.4	9.1	10.9	8.4	11.4	5	3.25	2.25	2.25	10.1	6.7
F11	2.95	5.65	7	10.9	9	11.8	3.55	2.95	2.95	2.95	10.3	8
F12	3	8.7	8.5	11.7	7.5	11.3	5	4	1	2	9.3	6
F13	1.8	7.5	9.1	11.7	4.5	11.3	5.9	4.5	1.2	3.1	9.9	7.5
Mean ranks	1.776923	7.126923	7.592308	10.47692	7.730769	10.76923	5.119231	4.730769	2.276923	2.907692	9.269231	8.223077

Table 15 Friedman test for the twelve algorithms (Dim = 100)

Function	STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F1	1	6	8	10.6	7	12	5	3	2	4	10.2	9.2
F2	1	6	8	7	12	11	5	3	2	4	9.7	9.3
F3	1	5	6.6	9.4	7.5	9.6	4	12	2	3	6.9	11
F4	1	7.4	5.2	10.1	8.1	11.4	4	9.4	2	3	5.8	10.6
F5	1	5.5	8	11.2	7	11.8	4.6	4.6	3.3	2	9	10
F6	2	6	8	10.8	7	12	5	4	1	3	10.2	9
F7	1	6	8	11.4	7	11.6	4.9	3.4	2.3	3.4	9	10
F8	1.1	10.9	6	11.9	5.4	5.5	9.4	1.9	4.8	3.3	8.5	9.3
F9	2.5	11.7	6.1	7.1	9.1	10.6	5	2.5	2.5	2.5	7.8	10.6
F10	2.2	6	8.2	11.4	8.6	10.9	5	3.4	2.2	2.2	9.6	8.3
F11	2.5	6	8.1	10.6	7	12	5	2.5	2.5	2.5	10	9.3
F12	2	6.5	7.5	11.6	7	11.4	5	3.9	1	3.1	9	10
F13	1.6	6	8	11.4	7	11.6	4.8	3	1.4	4.2	9	10
Mean ranks	1.530769	6.846154	7.361538	10.34615	7.669231	10.87692	5.130769	4.353846	2.230769	3.092308	8.823077	9.738462

Table 16 Friedman test for the twelve algorithms (Dim = 500)

Function	STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F1	1	6	7.1	9.6	7.9	11	5	3	2	4	9.4	12
F2	1	5.8	8	7	11.1	10	5.2	3	2	4	9	11.9
F3	1	6.3	5.8	10	8	9	4	11.7	2	3	5.9	11.3
F4	1	11	4	10.6	7.3	8.8	6	7.7	2	3	5.3	11.3
F5	1	6	7	10	8.5	11	5	3.1	2	3.9	8.5	12
F6	1	6	7.1	9.2	8.2	11	5	3	2	4	9.5	12
F7	1.1	6	7	10	8.2	11	5	3.1	2.4	3.4	8.8	12
F8	1.6	11	6.3	12	4.9	7.3	7.7	1.4	3.6	3.5	9.55	9.15
F9	2.5	9.4	7	6	9.8	10.8	5	2.5	2.5	2.5	8	12
F10	2.1	6	7	9.9	10.7	9.2	5	3.7	2.1	2.1	8.2	12
F11	2.5	5.9	7	9.7	8	11	5.1	2.5	2.5	2.5	9.3	12
F12	1	7	6	10	8.8	11	5	3	2	4	8.2	12
F13	1	6	7	10	8.8	11	4.5	3	2	4.5	8.2	12
Mean ranks	1.369231	7.107692	6.638462	9.538462	8.476923	10.16154	5.192308	3.9	2.238462	3.415385	8.296154	11.66538

Table 17 Friedman test for the twelve algorithms

Function	STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F14	4.85	11	4	8.9	6.35	4	9.1	7.4	7.7	4.75	6.15	3.8
F15	2.2	7.5	8.4	8.9	6	8.7	6.9	6.5	5.4	3.9	5	8.6
F16	2.1	10.6	4.95	11.8	10.1	2.45	9	8	6.3	4.3	3.95	4.45
F17	2.75	10.2	5.6	11.8	8	3.25	9.6	8.1	5.5	4.7	4.7	3.8
F18	3.75	10.9	6.2	10.5	8.6	4.3	9.6	9.9	5.35	2.45	3.85	2.6
F19	4.45	9.8	5	11.7	7.3	2.6	8.95	11.2	6.1	4.4	3.35	3.15
F20	4.9	7.2	8.5	11.3	6.8	4.6	6.7	8.1	6.7	4.55	3.45	5.2
F21	5	10.1	5.5	11.3	7.4	5.4	8.2	9.1	4.5	4.2	2.9	4.4
F22	4.7	9.7	5.6	11.5	7.4	3.05	6.95	8.7	5.9	4.95	4.35	5.2
F23	4.6	10.9	4.5	9.4	7.9	3.75	6.9	8.3	5.65	5.75	5.5	4.85
Mean ranks	3.93	9.79	5.825	10.71	7.585	4.21	8.19	8.53	5.91	4.395	4.32	4.605

4.4 Speed reducer design

This problem, as depicted in Fig. 20, is a renowned design challenge in mechanical systems. It revolves around optimizing the speed reducer, a vital component in gearbox applications. The objective is to minimize 11 constraints related to the weight of the speed reducer [51]. Among these constraints, seven are nonlinear, while the remaining are linear inequalities. The four crucial parameters influencing the optimization process include bending stress of the gear teeth, surface stress, transverse deflections of the shafts, and stresses in the shafts. To address this optimization problem, seven variables must be considered: face width (b), module of teeth (m), the number of teeth in the pinion (z), length of the first shaft between bearings (l_1), length of the second shaft between bearings (l_2), diameter of the first shaft (d_1), and diameter of the second shaft (d_2). The equation representing this problem is as follows:

$$\begin{aligned}
&\text{Minimize } f(b, m, z, l_1, l_2, d_1, d_2) \\
&= 0.7854x_1x_2^2\left(3.333x_3^2 + 14.9334x_3 - 43.0934\right) \\
&\quad - 1.508x_1\left(x_6^2 + x_7^2\right) \\
&\quad + 7.4777\left(x_6^3 + x_7^3\right) + 0.7854\left(x_4x_6^2 + x_5x_7^2\right)
\end{aligned}$$

$$\text{Subject to} \quad G_1 = \frac{27}{x_1x_2^2x_3} - 1 \le 0; G_2 = \frac{397.5}{x_1x_2^2x_3^2} - 1 \le 0;$$

$$G_3 = \frac{1.93x_4^3}{x_2x_6^4x_3} - 1 \le 0; G_4 = \frac{1.93x_5^3}{x_2x_7^4x_3} - 1 \le 0;$$

$$G_5 = \frac{\sqrt{\left(\frac{745x_4}{x_2x_3}\right)^2 + 16.9 \times 10^6}}{110x_6^3}$$

$$-1 \le 0; G_6 = \frac{\sqrt{\left(\frac{745x_5}{x_2x_3}\right)^2 + 157.5 \times 10^6}}{85x_7^3} - 1 \le 0;$$

$$G_7 = \frac{x_2x_3}{40} - 1 \le 0; G_8 = \frac{5x_2}{x_1} - 1 \le 0;$$

$$G_9 = \frac{x_1}{12x_2} - 1 \le 0;$$

$$G_{10} = \frac{1.5x_6 + 1.9}{x_4} - 1 \le 0; G_{11} = \frac{1.1x_7 + 1.9}{x_5} - 1 \le 0$$

Variable range: $2.6 \le x_1 \le 3.6; 0.7 \le x_2 \le 0.8; 17 \le x_3 \le 28; 7.3 \le x_4; x_5 \le 8.3; 2.9 \le x_6 \le 3.9; 5 \le x_7 \le 5.5;$

The achieved results that are presented in Table 28 demonstrate the STA algorithm finds a minimum cost design value in comparison with recent optimization methods including WOA, SSA, SCA, MFO, MVO, GWO, TSA, INFO, RUN, PSO, DE, SES [52], PSO [53], GSA [9], hybrid Harris Hawks sine cosine algorithm (hHHO-SCA) [54], multidisciplinary design optimization (MDO) [55], HS [56], adaptive firefly algorithm (AFA) [57], socio-behavioral simulation model (SBSM) [58], and aquila

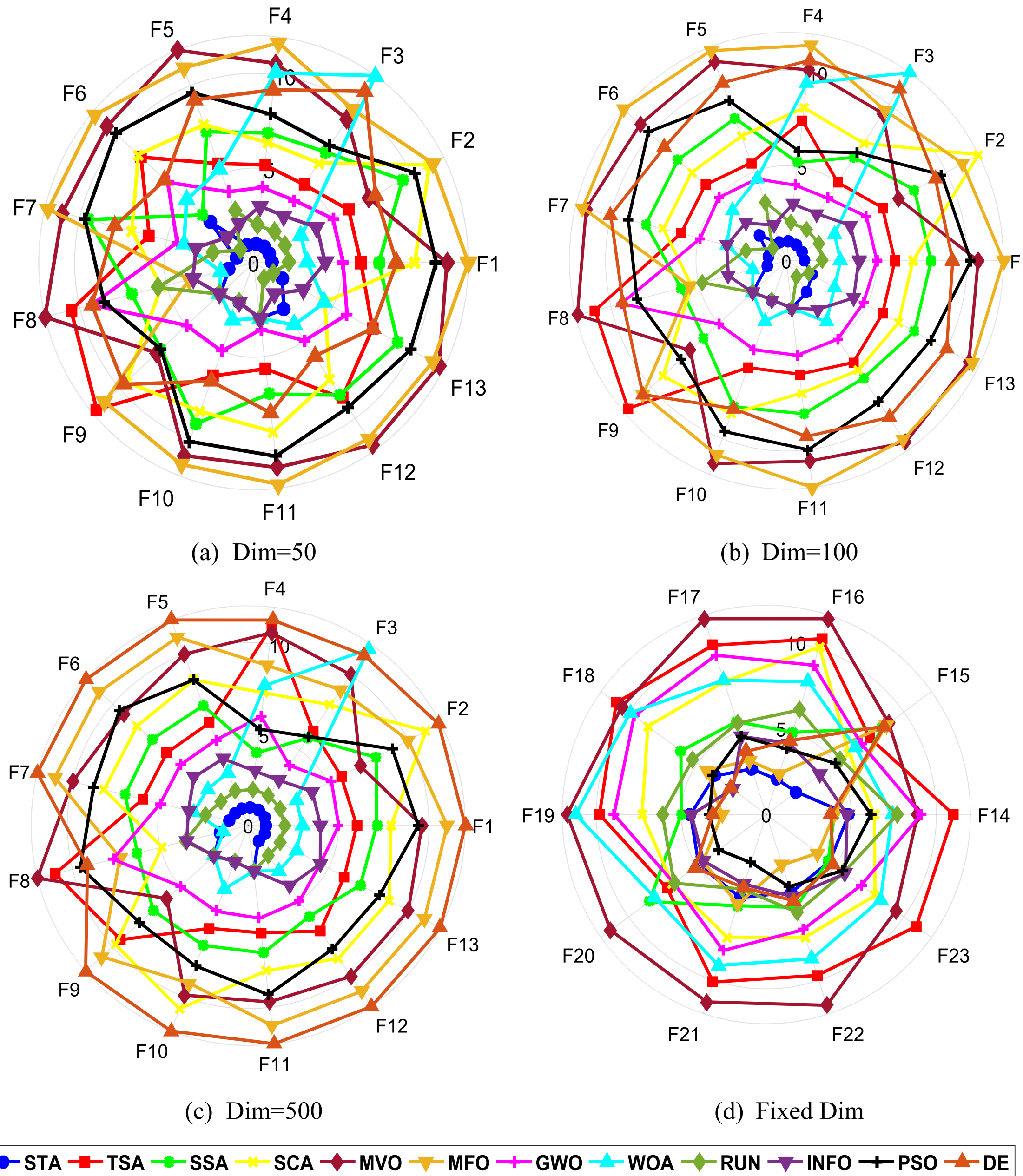


Fig. 9 Radar chart for the compared algorithms performance for 23 benchmark functions

optimizer (AO) [59]. Figure 21 showcases the convergence plots and box plots for all methods when tackling this problem. Notably, the proposed STA algorithm takes the lead in solving the speed reducer design challenge. In Table 29, you can find the statistical outcomes for the STA and the other chosen algorithms. A glance at Table 29 reveals that the STA outperformed all the selected methods. Consequently, it is evident that the proposed STA technique exhibits greater reliability for this engineering design problem.

4.5 Gear train design

As mentioned in Reference [60], the main goal of optimizing the gear train design is to approach a gear transmission ratio as closely as possible to 1/6.931. Let TA, TB, TD, and TF denote the number of teeth on gears A, B, D, and F, respectively. As each gear's number of teeth must be an integer within the range of 12–60, this optimization problem is converted into a constrained optimization problem with discrete variables. The formulation of the optimization problem is presented as follows:

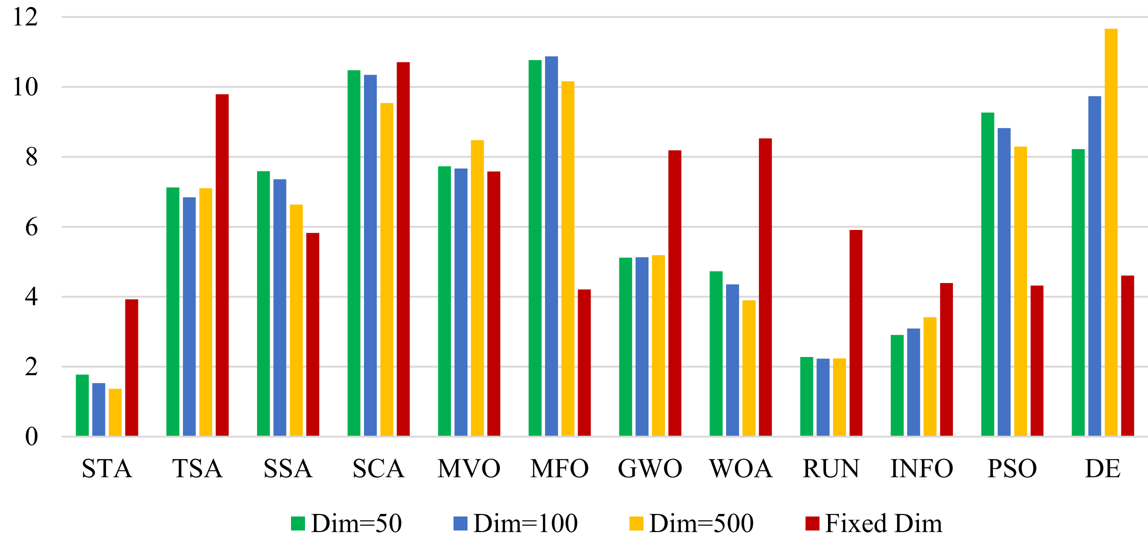


Fig. 10 Mean ranks achieved using Friedman test for 23 benchmark functions using several techniques

Table 18 Modern ten test functions from CEC2019

No	Functions	Dimension	Range	$f_{\min}$
CEC01	Storn's Chebyshev polynomial fitting	9	[− 8192, 8192]	1
CEC02	Inverse Hilbert matrix	16	[− 16,384, 16384]	1
CEC03	Lennard–Jones minimum energy cluster	18	[− 4, 4]	1
CEC04	Rastrigin's	10	[− 100,100]	1
CEC05	Griewangk's	10	[− 100,100]	1
CEC06	Weierstrass	10	[− 100,100]	1
CEC07	Modified Schwefel's	10	[− 100,100]	1
CEC08	Expanded Schaffer's F6	10	[− 100,100]	1
CEC09	Happy cat	10	[− 100,100]	1
CEC10	Ackley	10	[− 100,100]	1

$$\begin{aligned}&\text{Minimize} && f=\left(\frac{1}{6.931}-\frac{T_D T_B}{T_A T_F}\right)^2\\&\text{Variable range:} && 12\le T_A\le 60;\,12\le T_B\le 60;\,12\le T_D\le 60;\,12\le T_F\le 60;\end{aligned}\tag{1000}$$

Table 30 presents the optimal solutions found by STA and other techniques. This table reveals that the results obtained through the proposed STA technique are superior, with an optimal value of 0, while SCA yields the minimum results. The outcomes underscore the efficiency of the STA algorithm in solving the problem, with the design featuring the minimum fitness represented by $\vec{x}$ = [27.11492, 12.03429, 15.79045, 48.57386]. These experimental findings demonstrate that the proposed STA technique exhibits robust exploitation capabilities.

Figure 22 illustrates the convergence curves and box plots of the gear train design problem using the STA algorithm and other algorithms. The proposed STA algorithm is thoroughly assessed and compared with various optimization techniques, including WOA, SSA, SCA, MFO, MVO, GWO, INFO, RUN, PSO, DE, and TSA, all within the context of the gear train design problem. Table 31 provides the statistical outcomes for these algorithms, evaluating their performance based on metrics such as "best," "mean," "worst," and "Std." The STA algorithm delivers outstanding results, achieving a fitness value of 0, which signifies its capability to discover an optimal solution for the gear train design problem. Furthermore, it maintains an average fitness value of 0, indicating that, on average, it performs exceptionally well, closely approaching the optimal value. With a remarkably low standard deviation of 0, the STA algorithm demonstrates consistent performance and stability, consistently finding solutions that are close to the optimum. In summary, based on the statistical findings, the proposed STA algorithm exhibits exceptional performance when addressing the gear train design problem, surpassing the other evaluated algorithms. Its ability to consistently discover near-optimal solutions and its minimal variability make it a promising choice for similar constrained optimization problems.

Table 19 Numerical results for CEC 2019 benchmark functions by the STA technique and other well-known algorithms

Function		STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
CEC01	Best	43,622.41	41,425.08	66,415,704	1,134,504	8.45E + 08	1.38E + 08	50,387.31	5,290,653	37,688.56	33,479.17	3.93E + 08	1.3E + 10
	Mean	46,736.45	1.15E + 08	8.8E + 08	1.47E + 09	2.82E + 09	1.58E + 09	26,315,942	6.1E + 09	224,371.9	218,888.7	1.08E + 09	1.92E + 10
	Worst	50,525.77	9.16E + 08	2.42E + 09	4.49E + 09	5.4E + 09	3.85E + 09	1.09E + 08	1.96E + 10	858,516.1	816,513.2	2.32E + 09	3.18E + 10
	Std	2214.798	3.24E + 08	8E + 08	1.69E + 09	1.49E + 09	1.47E + 09	36,164,915	8.15E + 09	342,205.2	331,402.3	6.86E + 08	6.99E + 09
CEC02	Best	18.34286	18.35309	18.34303	18.38074	18.75652	18.34286	18.34317	18.34287	18.34286	18.34286	18.34286	18.34286
	Mean	18.34286	19.04584	18.34316	18.41653	18.93496	18.34286	18.34329	18.34299	18.34286	18.40996	18.40996	18.40996
	Worst	18.34286	19.71664	18.3433	18.46598	19.24245	18.34286	18.34343	18.34318	18.34287	18.75652	18.75652	18.75652
	Std	1.92E-09	0.518036	8.84E-05	0.027092	0.177624	0	8.59E-05	0.000103	3.64E-06	0.146511	0.146511	0.146511
CEC03	Best	13.7024	13.7024	13.7024	13.70241	13.7024	13.7024	13.7024	13.7024	13.7024	13.7024	13.7024	13.7024
	Mean	13.7024	13.70242	13.7024	13.70244	13.7024	13.7024	13.7024	13.7024	13.7024	13.7024	13.7024	13.70241
	Worst	13.7024	13.7025	13.7024	13.70252	13.7024	13.70241	13.7024	13.7024	13.7024	13.7024	13.7024	13.70241
	Std	3.8E-15	3.55E-05	1.9E-15	3.73E-05	6.06E-10	5.95E-07	1.03E-08	1.34E-09	6.17E-12	1.9E-15	1.9E-15	2.95E-06
CEC04	Best	57.73173	81.5409	13.93446	352.7773	16.57394	14.92942	25.5996	76.83637	50.7479	13.93446	23.88402	17.34498
	Mean	107.6644	1521.603	19.9042	714.3438	22.9174	49.25513	65.90642	149.1863	79.86647	56.21981	65.47301	49.72934
	Worst	166.1862	3499.279	26.86889	929.846	29.25836	125.3686	125.3686	322.1732	106.4645	125.3686	125.3686	125.3686
	Std	35.87244	1546.456	4.353187	194.8007	4.314048	47.47531	40.38744	78.22833	16.80202	44.87829	49.70768	46.96969
CEC05	Best	2.100903	2.48348	2.07386	2.921345	2.121544	2.105804	2.103819	2.126415	2.110833	2.066378	2.068869	2.073649
	Mean	2.275706	3.125763	2.274428	3.0394	2.283273	2.225399	2.172697	2.577741	2.245223	2.489082	2.217829	2.476063
	Worst	2.612948	4.082989	2.561093	3.161164	2.68216	2.469529	2.229679	3.012205	2.457652	3.060081	2.374036	3.060081
	Std	0.181432	0.497765	0.14507	0.068568	0.176481	0.122782	0.044175	0.321724	0.131509	0.461367	0.134922	0.470172
CEC06	Best	3.700687	9.021334	2.021989	9.745317	5.726581	4.056413	9.166096	6.936844	5.401056	6.25418	7.325756	7.858454
	Mean	6.966442	10.3951	5.910235	10.83007	7.331443	5.913596	10.74232	8.743194	8.118925	9.086115	9.339122	8.783592
	Worst	8.955334	11.2337	11.23916	11.37011	9.517057	9.819279	11.83582	10.38597	12.19503	11.0334	11.00086	9.35998
	Std	1.676464	0.891807	4.327551	0.572787	1.19549	1.788181	0.827094	1.046412	2.260056	1.512504	1.371517	0.551644
CEC07	Best	209.3351	283.5501	149.9107	150.2866	91.69736	31.04649	45.73065	99.32761	27.3364	111.9067	99.89545	26.73722
	Mean	328.8059	401.7173	288.5733	533.8364	248.4322	271.6748	394.7951	233.2405	202.053	246.2391	253.739	157.9107
	Worst	555.1042	693.1061	559.7542	720.5753	452.2623	384.1799	671.6189	420.1858	336.9728	487.9605	313.8561	336.9728
	Std	125.4988	148.9945	140.8175	169.6072	138.9276	129.2101	241.2132	107.4047	101.6348	124.2886	74.10955	91.22916
CEC08	Best	3.121673	5.362277	3.361663	4.479511	3.049914	4.171314	2.959204	4.747345	3.076661	3.764119	2.671945	4.287645
	Mean	4.533882	5.938339	4.882455	5.273074	4.325981	5.496916	4.012204	5.618261	4.207812	5.119399	4.391559	5.252786
	Worst	5.76691	6.535425	5.82449	5.818729	5.456317	6.255529	5.608334	6.237018	5.140331	6.212304	6.222693	6.057648
	Std	0.871468	0.444951	0.767731	0.520554	0.880478	0.683365	0.86905	0.512808	0.727455	0.800999	1.162107	0.555703
CEC09	Best	3.622603	5.03499	3.34267	15.05558	3.343156	3.35637	3.456414	4.40995	3.812663	3.35279	3.497789	3.403389
	Mean	4.300534	128.9178	4.403402	61.10513	3.354034	3.39608	4.541109	4.978612	4.703334	4.168103	4.263155	4.197348
	Worst	5.63545	363.4384	6.37797	226.6713	3.377815	3.476955	5.923895	6.308317	6.37797	6.37797	6.37797	6.37797
	Std	0.762632	171.1759	1.324221	68.20797	0.011825	0.035825	0.76071	0.67589	1.125221	1.369464	1.306224	1.346643
CEC10	Best	3.013515	21.27973	2.225259	21.27363	21.00983	20.99977	2.210882	21.0038	21.10492	20.99353	21.03478	21.0891
	Mean	18.74145	21.39531	18.78171	21.34458	21.01606	21.10525	18.90376	21.11745	21.36222	21.005	21.28683	21.16377
	Worst	21.00646	21.47131	21.3859	21.44923	21.02573	21.26351	21.39299	21.36276	21.49687	21.02088	21.44819	21.2391
	Std	6.355106	0.062043	6.692133	0.055831	0.00528	0.098934	6.745653	0.148718	0.112623	0.009107	0.153693	0.062043

Fig. 11 Convergence curves of all algorithms for CEC 2019 benchmark functions

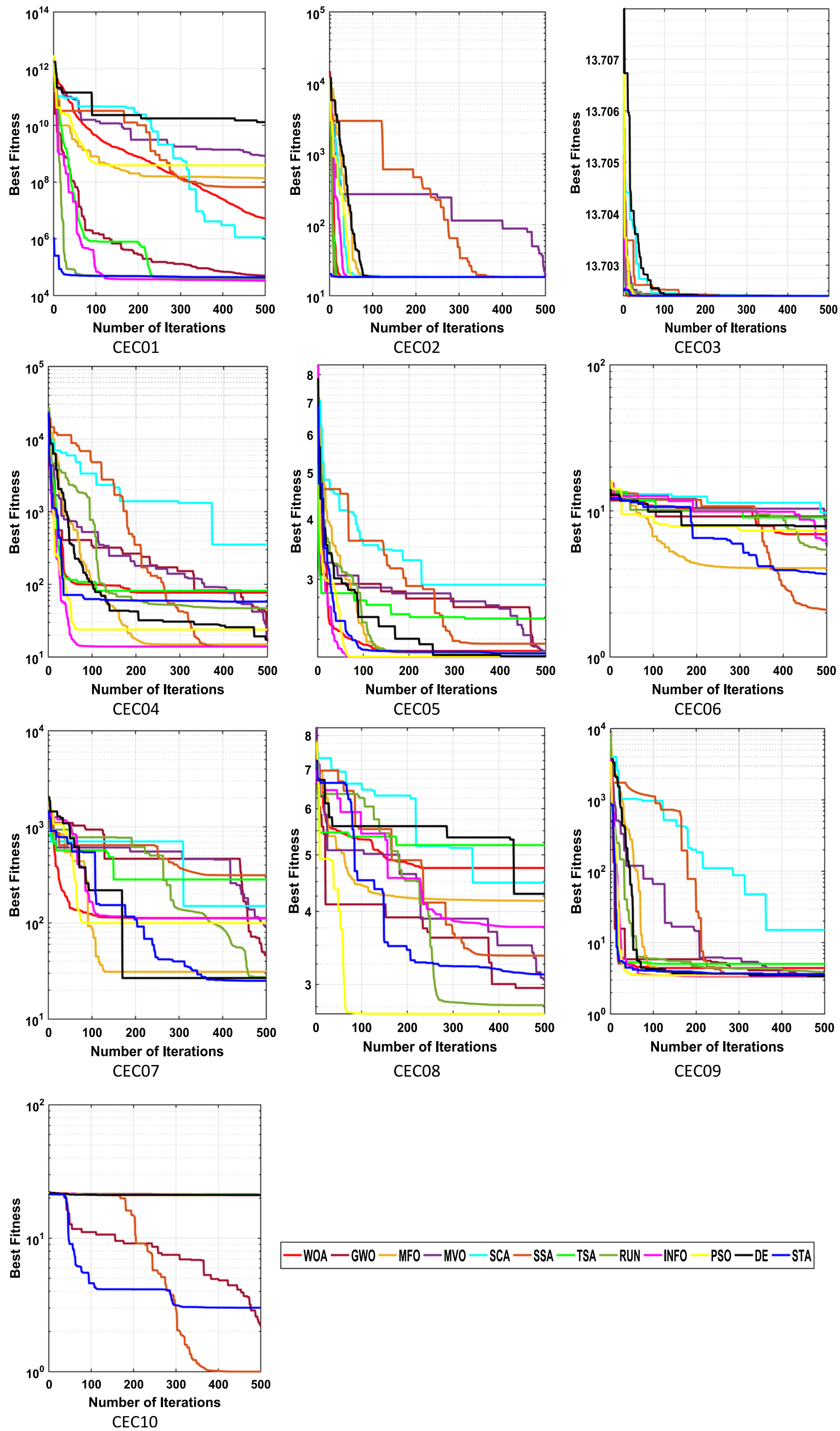


Table 20 Statistical results of the Wilcoxon rank-sum test for CEC 2019 benchmark functions

STA vs Fun	TSA		SSA		SCA		MVO		MFO	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
CEC01	6.50E-02	=	1.55E-04	+	1.55E-04	+	1.55E-04	+	1.55E-04	+
CEC02	1.55E-04	+	1.55E-04	+	1.55E-04	+	1.55E-04	−	1.55E-04	−
CEC03	1.55E-04	+	1.00E + 00	=	1.55E-04	+	1.55E-04	+	4.67E-01	=
CEC04	2.34E-01	=	1.55E-04	−	1.55E-04	+	1.55E-04	−	2.78E-02	−
CEC05	3.11E-04	+	9.59E-01	=	1.55E-04	+	7.21E-01	=	7.98E-01	=
CEC06	1.55E-04	+	5.74E-01	=	1.55E-04	+	1.00E + 00	=	1.95E-01	=
CEC07	1.85E-01	=	4.87E-01	=	2.07E-02	+	2.23E-01	=	1.00E + 00	=
CEC08	2.95E-03	+	3.28E-01	=	1.30E-01	=	9.59E-01	=	2.81E-02	+
CEC09	6.22E-04	+	5.56E-01	=	1.55E-04	+	1.55E-04	−	1.55E-04	−
CEC10	1.55E-04	+	3.82E-01	=	1.55E-04	+	1.55E-04	+	1.09E-03	+
WRST (+ / = / −)	7/3/0		2/7/1		9/1/0		3/4/3		3/4/3	

STA vs Fun	GWO		WOA		RUN		INFO		PSO		DE	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
CEC01	3.11E-04	+	1.55E-04	+	3.97E-01	=	8.60E-01	=	1.55E-04	+	1.55E-04	+
CEC02	1.55E-04	+	1.55E-04	+	1.55E-04	+	9.09E-02	=	9.93E-02	=	9.68E-02	=
CEC03	1.55E-04	+	1.55E-04	+	3.11E-04	+	1.00E + 00	=	1.00E + 00	=	1.55E-04	+
CEC04	8.22E-02	=	2.79E-01	=	1.05E-01	=	2.78E-02	−	1.08E-01	=	2.78E-02	−
CEC05	4.42E-01	=	4.99E-02	+	9.59E-01	=	6.45E-01	=	4.25E-01	=	8.78E-01	=
CEC06	1.55E-04	+	2.07E-02	+	5.74E-01	=	2.07E-02	+	1.04E-02	+	6.99E-03	+
CEC07	7.77E-01	=	1.85E-01	=	7.86E-02	=	2.23E-01	=	4.25E-01	=	2.95E-03	−
CEC08	3.28E-01	=	1.04E-02	+	5.05E-01	=	3.28E-01	=	7.98E-01	=	1.05E-01	=
CEC09	5.74E-01	=	1.30E-01	=	2.67E-01	=	1.87E-01	=	1.25E-01	=	1.01E-01	=
CEC10	1.04E-02	+	1.09E-03	+	1.55E-04	+	1.05E-01	=	1.55E-04	+	1.55E-04	+
WRST (+ / = / −)	5/5/0		7/3/0		3/7/0		1/8/1		3/7/0		4/4/2	

Table 21 Friedman test using the twelve algorithms for CEC 2019 benchmark functions

Function	STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
F14	2.4375	3.375	7.875	7.875	9.75	8.25	4.9375	8.875	2.0625	2.5625	8.25	11.75
F15	4.25	11	7.375	9.375	11.5	1.9375	8	6.375	5.25	4.0625	4.3125	4.5625
F16	3.125	10.5	2.8125	11.75	7.625	4.5625	8.375	7.25	5.75	2.8125	2.8125	10.625
F17	8.25	9.75	2.125	11.5	2.5	4.75	6.625	9.25	7.25	5.1875	6.25	4.5625
F18	5	10.625	5.75	11.125	5.75	5.25	3.625	8.125	5.125	6.625	4.875	6.125
F19	3.5	9.875	4.75	10.25	3.875	2.625	10.25	6	5.25	7.25	7.875	6.5
F20	7	9.375	6.375	10.5	5.375	6.625	8	5.125	4.8125	5.25	6	3.5625
F21	4.875	10.5	5.875	7.75	4.5	9	3.125	9.375	3.75	7	4.75	7.5
F22	7.125	10.875	4.75	11.625	1.375	2.875	7.25	8.125	7.125	5.125	6	5.75
F23	1.875	10.625	4.75	9.5	3.875	5.375	7.875	5.875	10	2.625	8.75	6.875
Mean ranks	4.74375	9.65	5.24375	10.125	5.6125	5.125	6.80625	7.4375	5.6375	4.85	5.9875	6.78125

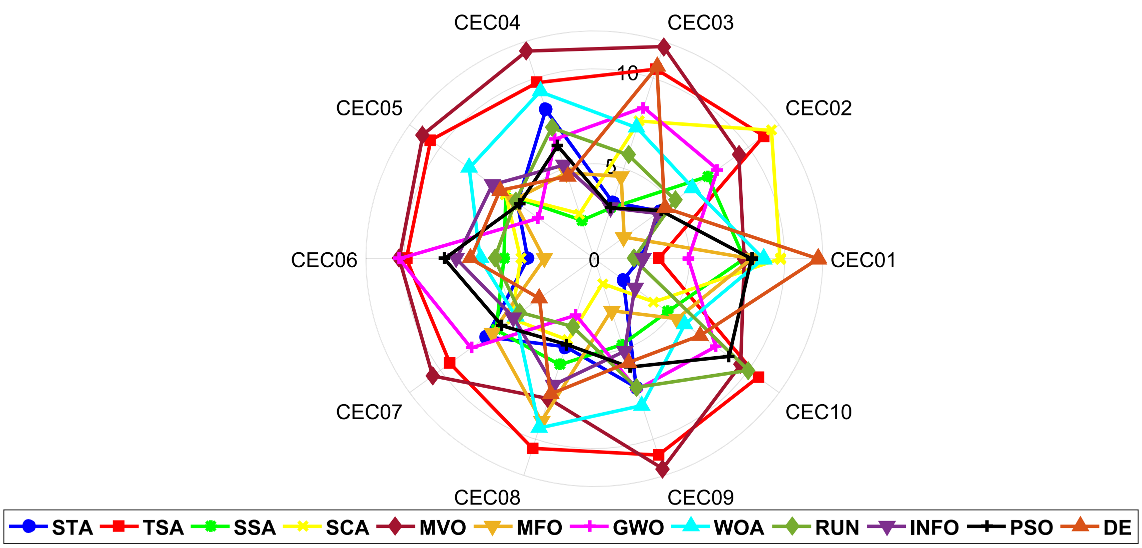


Fig. 12 Radar chart for the compared algorithms performance for CEC 2019 benchmark functions

4.5.1 Wilcoxon's rank test results

The proposed STA has been evaluated against several other optimization algorithms, namely TSA, SSA, SCA, MVO, MFO, GWO, RUN, INFO, PSO, DE, and WOA, on a set of real-world constrained engineering design problems. The results of this evaluation are summarized in Table 32, which displays the p-values and the corresponding winners for each comparison between STA and the other algorithms for various optimization problems including BTD, CBD, WBD, SRD, and GTD. In summary, the proposed STA algorithm demonstrates competitive performance compared to other optimization algorithms across various engineering design problems. It outperforms or is statistically equivalent to several algorithms for different problems.

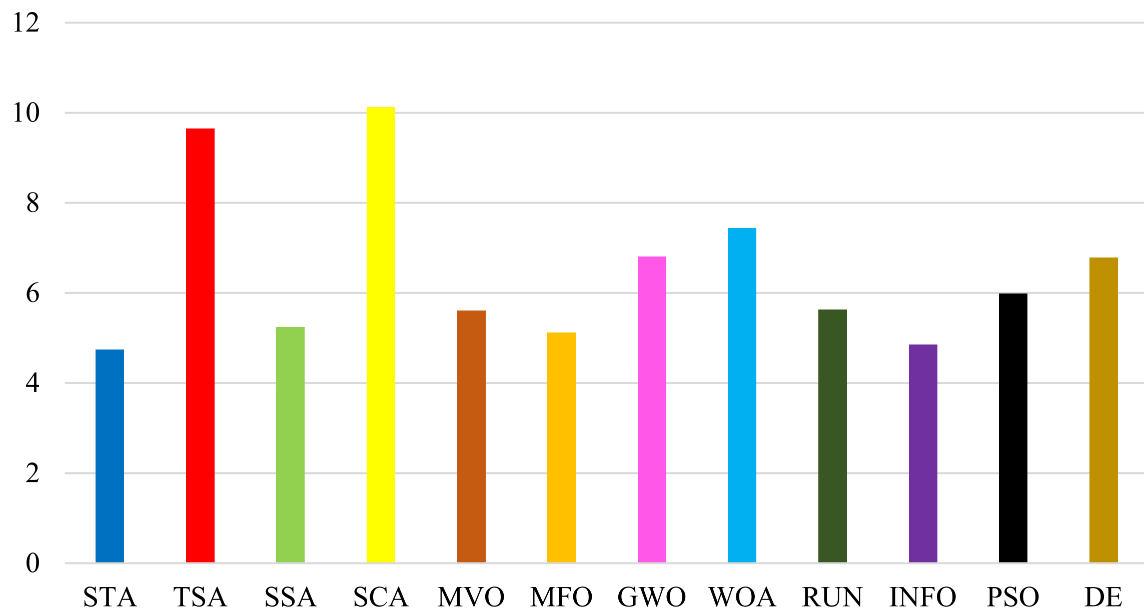


Fig. 13 Mean ranks achieved using Friedman test for CEC 2019 benchmark functions using several techniques

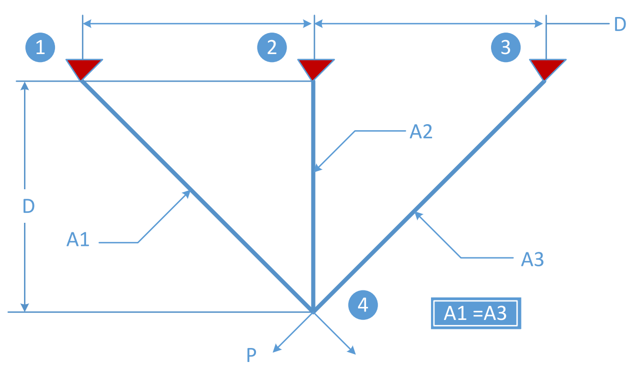


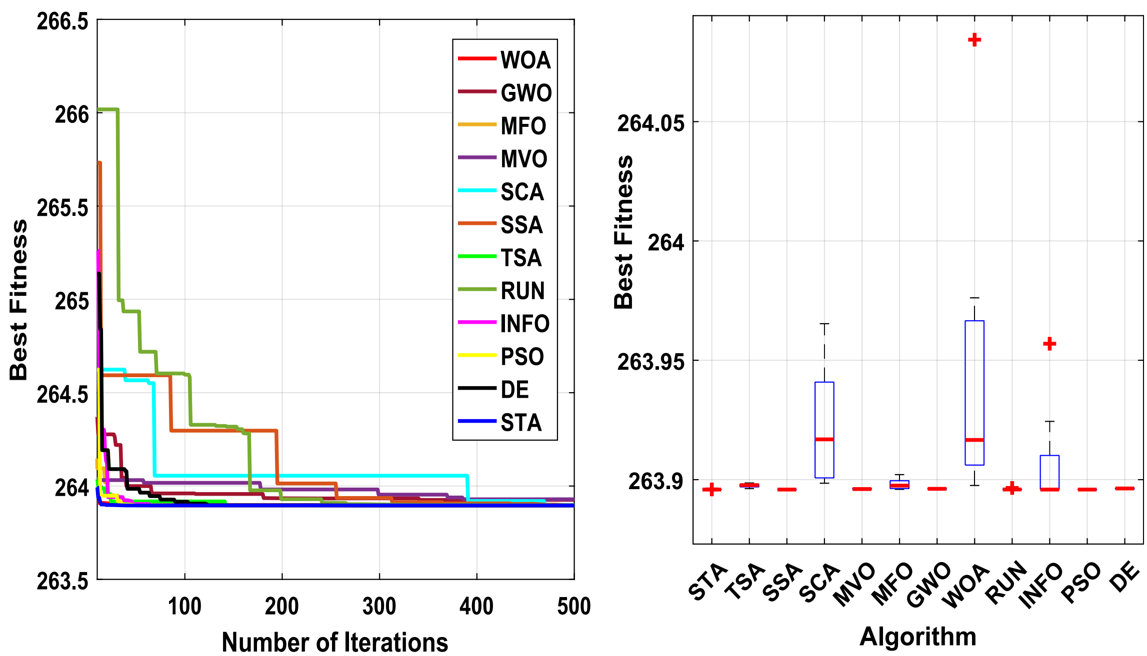
Fig. 14 Schematic of three-bar truss problem

4.5.2 Friedman's rank test results

Table 33 presents the results of the Friedman test, which assessed the performance of eight different optimization algorithms across a set of real-world constrained engineering design problems. For the BTD problem, STA achieved a mean rank of 2.6, indicating its superior performance compared to the other algorithms, with WOA having the highest mean rank of 11.6. In the case of the CBD problem, STA obtained a mean rank of 3.6, making it one of the top-performing algorithms after that INFO, RUN and SSA, while SCA had the highest mean rank of 11.6. Regarding the WBD problem, STA achieved a mean rank of 2.4, placing it among the top-performing algorithms. MFO had the lowest mean rank of 1.35, indicating its strong performance. For the SRD problem, STA's mean rank was 2.8, signifying competitive performance, while SCA had the highest mean rank of 11.9. In the GTD problem, STA obtained a mean rank of 3.15, demonstrating strong performance, while SCA had the highest mean rank of 11.4. Overall, the mean ranks for each algorithm across different problems were calculated. STA had a mean rank of 2.91, making it one of the top-performing algorithms in this evaluation, while SCA had the highest mean rank of 11.26. These results suggest that STA consistently performed well across multiple engineering design problems during compared to the other techniques.

Table 22 Results for the three-bar truss problem

Algorithms	Optimum values for variables		Optimal cost
	x_1	x_2	
STA	0.78867514354	0.40824825645	263.8958429412
TSA	0.788303894793	0.409302223077	263.8962341653
SSA	0.7886705766	0.4082611737	263.8958429566
SCA	0.790289750	0.403708161	263.898512530
MVO	0.78869593269	0.40818939236	263.8958687956
MFO	0.78886299861	0.40771718127	263.8958688599
GWO	0.7884720971	0.4088228569	263.8958733906
WOA	0.78715946310	0.41255221852	263.89753998011
RUN	0.788611292	0.408428885	263.8958459
INFO	0.78867513	0.408248295	263.8958429
PSO	0.788670295	0.408261971	263.895843
DE	0.788685459	0.408219107	263.8958451
Ray and sain [33]	0.795	0.395	264.3
AAA [34]	0.788735	0.408078	263.8959
MBA [35]	0.788565	0.40856	263.8959
DEDS [36]	0.788675	0.408248	263.8958
GOA [37]	0.788898	0.40762	263.8959
PSO-DE [38]	0.788675	0.408248	263.8958
CS [39]	0.78867	0.40902	263.9716



(a) Convergence curves (b) Boxplots

Fig. 15 Convergence curves and boxplots of the studied techniques for three-bar truss problem

Table 23 Statistical results for the three-bar truss problem

Function	STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
Best	263.8958	263.8962	263.8958	263.8985	263.8959	263.8959	263.8959	263.8975	263.8958	263.8958	263.8958	263.8958
Mean	263.8958	263.8975	263.8959	263.9195	263.8962	263.8977	263.8961	263.9675	263.8959	263.9048	263.8959	263.8963
Worst	263.8959	263.8986	263.8959	263.9653	263.8966	263.9021	263.8962	264.1226	263.8965	263.9569	263.896	263.8966
Std	1.51E-05	0.000693	2.59E-05	0.023031	0.000253	0.002142	0.000121	0.079033	0.00021	0.020397	4.3E-05	0.000231

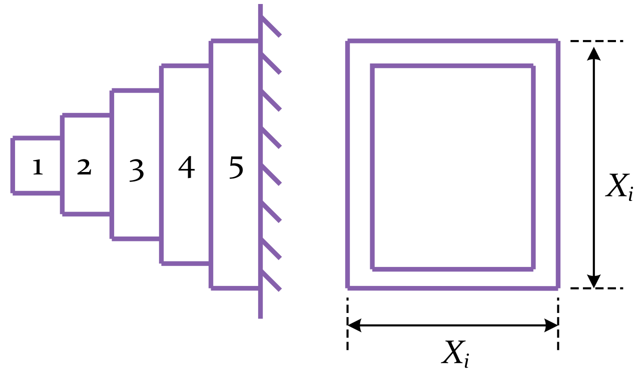


Fig. 16 Schematic of cantilever beam design problem

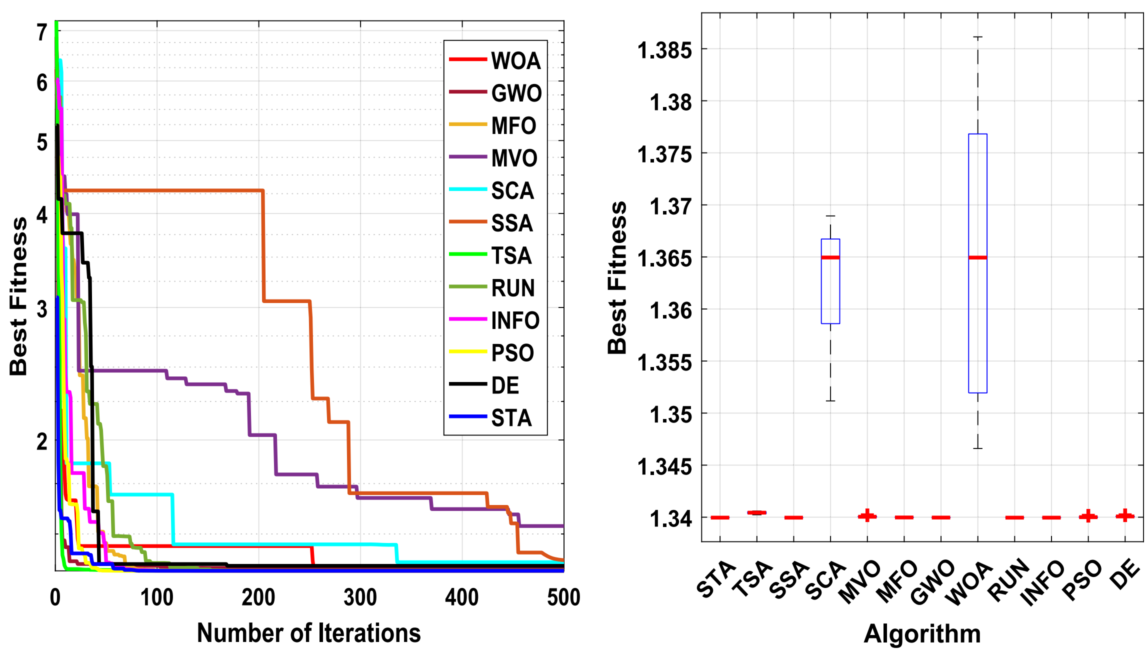
Conversely, in Fig. 23, a radar chart provides a visual representation of the ranks achieved by all the compared algorithms. This chart offers a succinct and intuitive way to observe how each algorithm fares in comparison to others when dealing with real-world constrained engineering design problems. The radar chart's multi-dimensional approach allows for a rapid assessment of each algorithm's strengths and weaknesses in addressing the diverse array of problems encountered in real-world constrained engineering design. Furthermore, Fig. 24 displays the Mean ranks acquired through the Friedman test for real-world constrained engineering design problems. These figures unambiguously illustrate that the STA technique secures the most favorable average rank, indicating its top position among all the techniques.

5 Conclusion

His article presents the development of a novel metaheuristic technique called the supercell thunderstorm algorithm (STA). The STA algorithm is inspired by the behavior of supercell thunderstorms in nature, which are severe thunderstorms known for their rotating updrafts, strong wind shear, and potential for tornado formation. The algorithm is designed with three main components: spiral motion, formation, and jet stream, to optimize procedures. To evaluate the effectiveness of the STA algorithm, it is rigorously tested on a set of 23 benchmark functions with varying dimensions. The evaluation focuses on the algorithm's ability to explore and exploit solution spaces effectively, thus avoiding local optima. The initial results

Table 24 Attained results of the cantilever beam design problem

Algorithms	Optimal values for variables					Optimum cost
	x_1	x_2	x_3	x_4	x_5	
STA	6.014644	5.310063	4.494105	3.501842	2.153006	1.339956431
TSA	5.95913658	5.31046273	4.47525089	3.53800374	2.19482723	1.34020730
SSA	6.0170359	5.3087936	4.4943621	3.5010606	2.1524080	1.3399564
SCA	6.15679718	5.27683125	4.36523884	3.69294174	2.16157645	1.35117125
MVO	6.0194905	5.3204335	4.4967766	3.4949515	2.1423633	1.3399786
MFO	6.01966309	5.32538293	4.48700697	3.49353430	2.14825345	1.33996766
GWO	6.0203108	5.3015291	4.4956415	3.5024256	2.1537854	1.3399584
WOA	5.96412747	5.35766213	4.91330380	3.31761582	2.02746203	1.34660269
RUN	6.0162641	5.30899182	4.493281909	3.50266012	2.1524633	1.339956458
INFO	6.016183999	5.30897862	4.4943691	3.5013780	2.15274992	1.33995636
PSO	6.01396675	5.31876938	4.4823418	3.51008259	2.1486565	1.3399662
DE	6.000452	5.328834	4.521241	3.489147	2.135171	1.34003
GCA_I [41]	6.01	5.304	4.49	3.498	2.15	1.34
ALO [14]	6.01812	5.31142	4.48836	3.49751	2.158329	1.33995
GCA_II [41]	6.01	5.3	4.49	3.49	2.15	1.34
MMA [41]	6.01	5.3	4.49	3.49	2.15	1.34
SOS [42]	6.01878	5.30344	4.49587	3.49896	2.15564	1.33996
ISMA [43]	6.017757	5.310892	4.493758	3.501106	2.150159	1.33996



(a) Convergence curves (b) Boxplots

Fig. 17 Convergence curves and boxplots for cantilever beam design problem using the studied methods

Table 25 Numerical results for the cantilever beam design problem

Function	STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
Best	1.339956	1.340207	1.339956	1.351171	1.339979	1.339968	1.339958	1.346603	1.339956	1.339956	1.339966	1.34003
Mean	1.33996	1.340416	1.339957	1.362864	1.340054	1.339991	1.339967	1.36271	1.339957	1.339957	1.340009	1.340085
Worst	1.339967	1.340593	1.339957	1.36894	1.340186	1.340032	1.339973	1.386139	1.339957	1.339957	1.34014	1.340186
Std	3.8E-06	0.000136	3.78E-07	0.005474	5.73E-05	2.46E-05	4.84E-06	0.013973	2.96E-07	2.93E-07	5.04E-05	5.69E-05

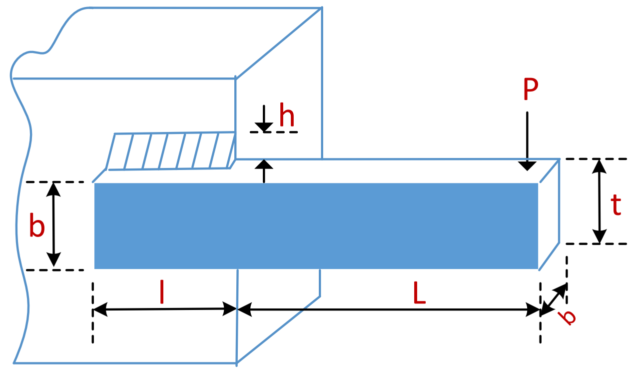


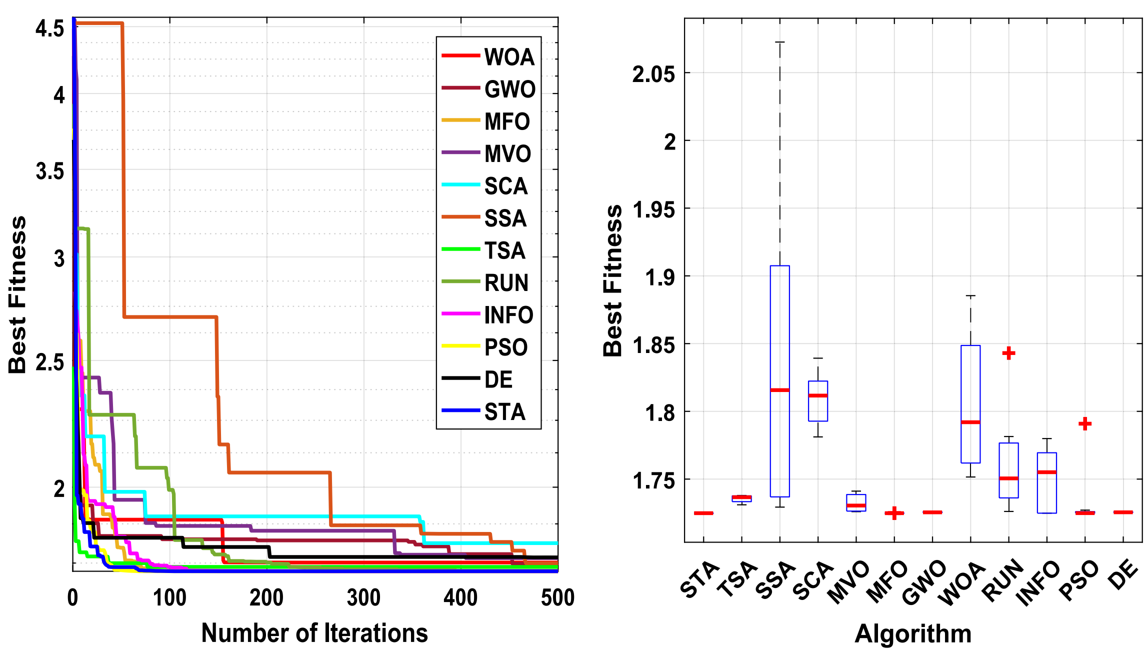
Fig. 18 Schematic of welded beam design problem

clearly demonstrate that the STA algorithm outperforms well-known algorithms such as the tunicate swarm algorithm (TSA), salp swarm algorithm (SSA), sine cosine algorithm (SCA), multi-verse optimizer (MVO), moth flame optimization (MFO), grey wolf optimizer (GWO), whale optimization algorithm (WOA), Runge–Kutta optimizer (RUN), weighted mean of vectors (INFO), particle swarm optimization (PSO), and differential evolution (DE). Across all 31 benchmark functions, the STA algorithm consistently exhibits superior performance in terms of exploration, exploitation, avoidance of local optima, and convergence characteristics. Statistical tests are conducted to confirm the STA algorithm's superiority over other metaheuristics. Furthermore, the study validates the effectiveness of the STA algorithm in solving complex composition functions by employing the CEC2019 functions. The results demonstrate that STA efficiently discovers optimal solutions for a majority of benchmark functions while keeping a balanced approach between exploration and exploitation. As well as the benchmark functions, the STA algorithm is applied to five real-world constrained engineering design issues, including three-bar truss design (BTD), cantilever beam design (CBD), welded beam design (WBD), speed reducer design (SRD), and gear train design (GTD). These practical applications highlight the high-performance capabilities of the STA algorithm in navigating unexplored search spaces.

The remarkable performance demonstrated by the STA algorithm, as discussed earlier, opens up a broad range of promising research directions for future exploration. These avenues encompass the application of STA to various problem domains, including PV parameter extraction, smart home challenges, industrial and engineering issues, neural networks, image processing tasks, text and data mining problems, big data challenges, signal denoising, resource management applications, network optimization,

Table 26 Results of the welded beam design problem

Methods	Optimum values for variables				Optimal cost
	h	*l*	*t*	*b*	
STA	0.20572964	3.47048867	9.03662391	0.20572964	1.72485231
TSA	0.20576363	3.48760424	9.03504322	0.20626527	1.73103528
SSA	0.20251079	3.54113817	9.03662389	0.20572965	1.72933413
SCA	0.204143709	3.562728278	9.317920839	0.205398635	1.781148515
MVO	0.20546654	3.48127685	9.03693041	0.20573182	1.72597517
MFO	0.20572964	3.47048867	9.03662391	0.20572964	1.72485231
GWO	0.205692461	3.472186880	9.038687658	0.205722209	1.725325363
WOA	0.201062681	3.523553491	9.164860059	0.206328680	1.751559804
RUN	0.204799353	3.49062759	9.036623764	0.20572965	1.726122525
INFO	0.20572964	3.470488666	9.03662391	0.20572964	1.72485231
PSO	0.20572964	3.470488662	9.036623905	0.20572964	1.72485231
DE	0.20567864	3.47413978	9.03698296	0.20573164	1.72534632
GOA [37]	0.182129	3.856979	10	0.202376	1.87995
RO [45]	0.203687	3.528467	9.004233	0.207241	1.735344
ECDE [46]	0.203137	3.542998	9.033498	0.206179	1.733461
GA [47]	0.2489	6.173	8.1789	0.2533	2.43
HS [48]	0.2442	6.2231	8.2915	0.24	2.3807
CPSO [49]	0.202369	3.544214	9.04821	0.205723	1.72802
MPA [21]	0.205728	3.470509	9.036624	0.20573	1.724853



(a) Convergence curves (b) Boxplots

Fig. 19 Convergence curves and boxplots using all algorithms for welded beam design issue

other test functions, feature selection, image segmentation, task scheduling, and many more.

Additionally, there is ample opportunity to extend the applicability of STA to real-world scenarios involving discrete, binary, and multi-objective optimization issues.

Table 27 Numerical results for the welded beam design problem

Algorithm	Best	Mean	Worst	Std
STA	1.724852	1.724853	1.724864	3.58E-06
TSA	1.731035	1.735537	1.738094	0.002721
SSA	1.729334	1.880685	2.175366	0.152304
SCA	1.781149	1.807943	1.839232	0.018036
MVO	1.725975	1.731615	1.741135	0.005823
MFO	1.724852	1.724856	1.724892	1.26E-05
GWO	1.725325	1.725536	1.726064	0.000237
WOA	1.75156	1.804441	1.885408	0.048632
RUN	1.726123	1.759784	1.843008	0.033599
INFO	1.724852	1.757138	1.843008	0.03708
PSO	1.724852	1.732419	1.790889	0.020672
DE	1.725346	1.725588	1.726064	0.000235
Rao-1 [50]	1.724852	1.724852	1.724852	2.62
Rao-2 [50]	1.724852	1.724852	1.724852	9.83E-04
Rao-3 [50]	1.724852	1.724852	1.724852	2.06E-03
FISA [50]	1.724852	1.724852	1.724852	5.93E-05
MPA [21]	1.724853	1.724861	1.724873	6.41E-06

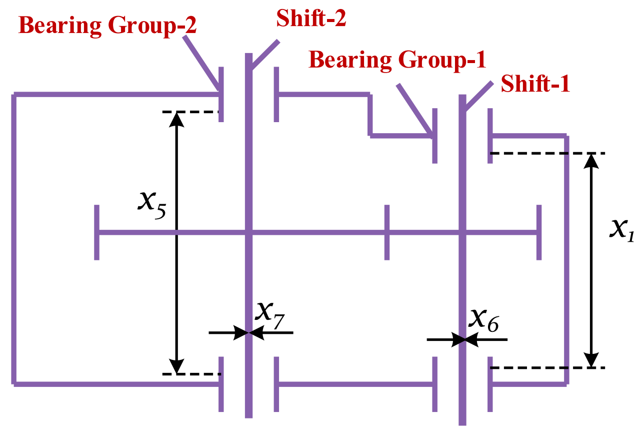


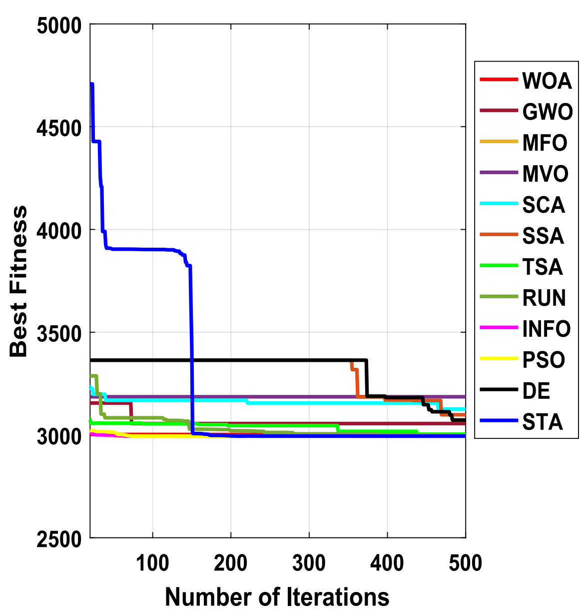
Fig. 20 Schematic of speed reducer design problem

To further enhance the performance of the STA algorithm, it is worth investigating approaches such as incorporating disruptive elements, mutation mechanisms, Levy flight dynamics, and other stochastic components. These explorations can be conducted within the context of both local and global search strategies, while also considering the integration of additional evolutionary operators.

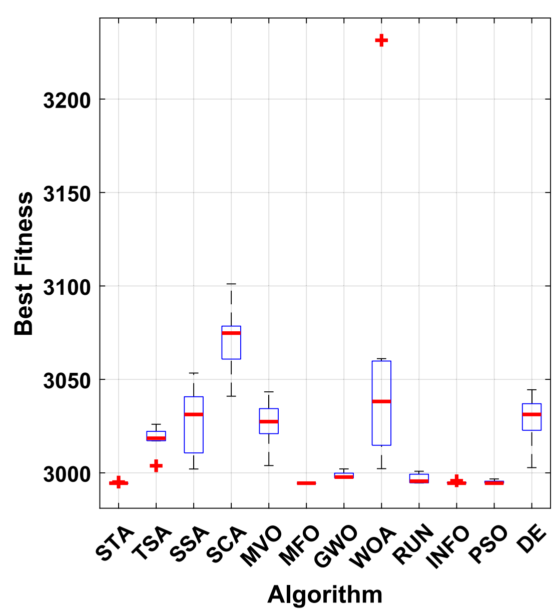
Furthermore, it may be worthwhile to explore the development of a discrete version of the STA algorithm to address discrete optimization challenges effectively. These avenues of future research hold great promise for unlocking additional potential in the application and advancement of the STA algorithm.

Table 28 Comparison results for speed reducer design problem

Algorithms	Optimum values for variables							Optimal cost
	b	m	p	l_1	l_2	d_1	d_2	
STA	3.5	0.7	17	7.3	7.71532	3.350215	5.286654	2994.471
TSA	3.51003	0.7	17	7.567897	7.72085	3.355346	5.28914	3003.789
SSA	3.50101	0.7	17	7.582481	7.920558	3.350826	5.286725	3002.064
SCA	3.53224	0.706	17	8.3	7.996289	3.407131	5.302415	3074.796
MVO	3.50672	0.7	17	7.3	8.005923	3.350915	5.28701	3003.895
MFO	3.5	0.7	17	7.3	7.71532	3.350215	5.286654	2994.471
GWO	3.50034	0.7	17	7.350353	7.757172	3.353536	5.287135	2997.119
WOA	3.50094	0.7	17	7.589163	7.906494	3.350765	5.287461	3002.241
RUN	3.5	0.7	17	7.3	7.719811	3.35023	5.286656	2994.575
INFO	3.5	0.7	17	7.3	7.71532	3.350215	5.286654	2994.471
PSO	3.5	0.7	17	7.3	7.71532	3.350215	5.286654	2994.472
DE	3.504446	0.7	17	7.511366	7.858522	3.356191	5.286703	3002.787
SES [52]	3.50616	0.7008	17	7.460181	7.962143	3.3629	5.308949	3025.005
PSO [53]	3.5001	0.7	17.0002	7.5177	7.7832	3.3508	5.2867	3145.922
GSA [9]	3.6	0.7	17	8.3	7.8	3.369658	5.289224	3051.12
hHHO-SCA [54]	3.50612	0.7	17	7.3	7.99141	3.452569	5.286749	3029.873
MDO [55]	3.5	0.7	17	7.3	7.670396	3.542421	5.245814	3019.583
HS [56]	3.52012	0.7	17	8.37	7.8	3.36697	5.288719	3029.002
AFA [57]	3.50749	0.7001	17	7.719674	8.080854	3.351512	5.287051	3010.137
SBSM [58]	3.50612	0.7	17	7.549126	7.85933	3.365576	5.289773	3008.08
AO [59]	3.5021	0.7	17	7.3099	7.7476	3.3641	5.2994	3007.733



(a) Convergence curves



(b) Boxplots

Fig. 21 Convergence curves and boxplots using all algorithms for speed reducer design problem

Table 29 Numerical results for speed reducer design problem

Function	STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
Best	2994.471	3003.789	3002.064	3041.012	3003.895	2994.471	2997.119	3002.241	2994.575	2994.471	2994.472	3002.787
Mean	2994.53	3019.463	3023.785	3070.708	3023.272	2994.471	2999.108	3050.551	2996.981	2994.814	2994.927	3030.501
Worst	2995.058	3026.007	3053.4	3101.098	3043.344	2994.471	3002.097	3231.458	3000.866	2995.76	2996.736	3044.471
Std	0.185479	6.461037	18.06583	16.18592	13.48128	3.71E-13	2.028856	66.90673	2.343144	0.565308	0.868838	12.93881

Table 30 Results of the gear train design problem

Algorithms	Optimal values for variables				Optimum cost
	T_A	T_B	T_D	T_F	
STA	27.11492	12.03429	15.79045	48.57386	0
TSA	60	16.10947	23.06112	42.91472	5.58E-17
SSA	51.03928	13.02365	14.41983	25.5026	1.84E-26
SCA	59.624	19.62691	23.4388	53.47629	6.86E-14
MVO	30.21696	20.32619	12	55.94772	1.69E-17
MFO	44.87759	12	13.24268	24.54276	0
GWO	53.72381	15.38978	21.10562	41.9044	1.33E-17
WOA	57.8992	28.6439	16.67487	57.17653	0
RUN	53.48283	12.0551	12.01219	18.7661	0
INFO	42.61166	15.89775	15.01064	38.81523	0
PSO	54.90092	18.17824	21.30102	48.88421	0
DE	55.48742	39.1983	13.22674	64.7623	5.75E-14

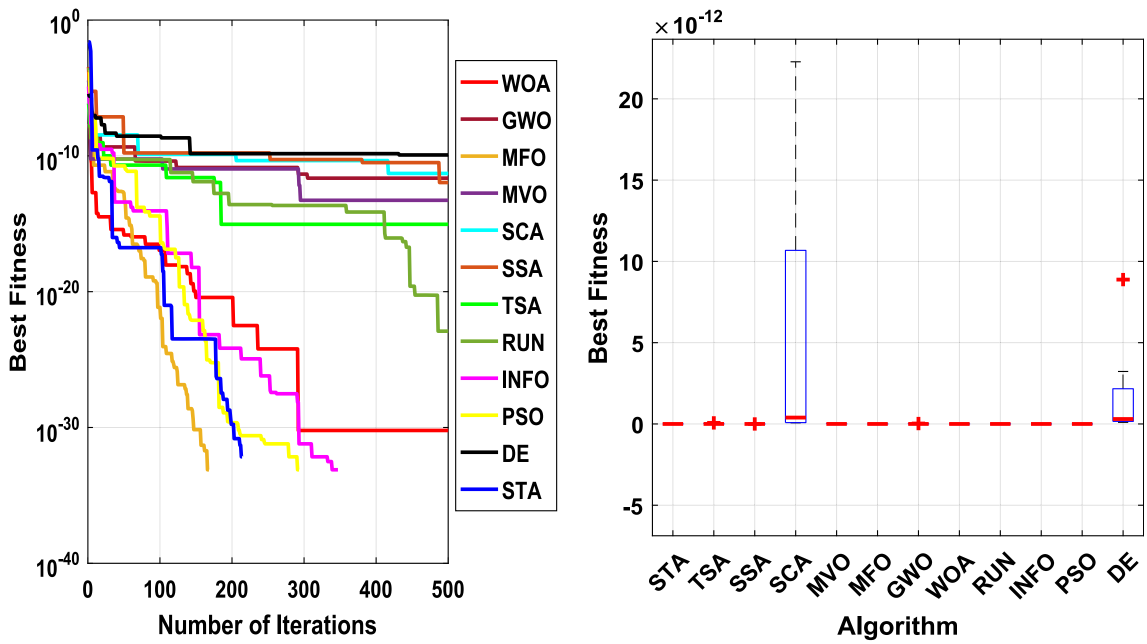


Fig. 22 Convergence curves and boxplots of all algorithms for gear train design problem

Table 31 Statistical results for gear train design problem

Function	STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
Best	0	5.58E-17	1.84E-26	6.86E-14	1.69E-17	0	1.33E-17	0	0	0	0	5.75E-14
Mean	0	9E-15	7.48E-22	4.64E-12	6.98E-15	0	1.28E-14	0	8.79E-18	0	0	2.17E-12
Worst	0	6.59E-14	4.62E-21	2.23E-11	3.13E-14	0	7.82E-14	0	3.45E-17	0	0	8.88E-12
Std	0	2.03E-14	1.41E-21	8.16E-12	1.01E-14	0	2.44E-14	0	1.31E-17	0	0	3.3E-12

Table 32 Statistical results of the Wilcoxon rank-sum test for real-world constrained engineering design problems

STA vs Fun	TSA		SSA		SCA		MVO		MFO	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
BTD	1.83E-04	+	2.73E-01	=	1.83E-04	+	2.46E-04	+	2.46E-04	+
CBD	1.83E-04	+	3.12E-02	−	1.83E-04	+	1.83E-04	+	1.83E-04	+
WBD	1.83E-04	+	1.83E-04	+	1.83E-04	+	1.83E-04	−	2.04E-03	−
SRD	1.82E-04	+	1.82E-04	+	1.82E-04	+	1.82E-04	−	1.40E-04	−
GTD	6.39E-05	+	6.39E-05	+	6.39E-05	+	6.39E-05	+		=
WRST (+ / = / −)	5/0/0		3/1/1		5/0/0		3/0/2		2/1/2	

STA vs Fun	GWO		WOA		RUN		INFO		PSO		DE	
	P	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner	*P*	Winner
BTD	2.46E-04	+	1.83E-04	+	1.40E-02	+	2.44E-02	−	2.12E-01	=	7.55E-04	+
CBD	5.80E-03	+	1.83E-04	+	3.12E-02	−	9.11E-03	−	2.46E-04	+	1.79E-04	+
WBD	1.83E-04	+	1.83E-04	+	1.83E-04	+	4.71E-01	+	1.01E-03	+	1.81E-04	+
SRD	1.82E-04	+	1.82E-04	+	4.37E-04	+	1.37E-01	+	1.70E-03	+	1.78E-04	+
GTD	6.39E-05	+		=	2.21E-03	+		=		=	6.39E-05	+
WRST (+ / = / −)	5/0/0		4/1/0		4/0/1		2/1/2		3/2/0		5/0/0	

Table 33 Friedman test using the twelve algorithms for real-world constrained engineering design problems

Function	STA	TSA	SSA	SCA	MVO	MFO	GWO	WOA	RUN	INFO	PSO	DE
BTD	2.6	9.2	3.3	10.9	6.9	8.3	6.5	11.6	4.6	3.1	3.9	7.1
CBD	3.6	10	2.5	11.6	8.1	6.6	5	11.4	2.4	1.8	6.5	8.5
WBD	2.4	7.2	10.6	10.5	6.8	1.35	4.6	10.4	9	6.05	4.4	4.7
SRD	2.8	8.5	8.6	11.9	8.8	1.35	6	9.6	4.9	2.25	3.8	9.5
GTD	3.15	8.9	6.3	11.4	9.1	3.15	9.1	3.15	5.95	3.15	3.15	11.5
Mean ranks	2.91	8.76	6.26	11.26	7.94	4.15	6.24	9.23	5.37	3.27	4.35	8.26

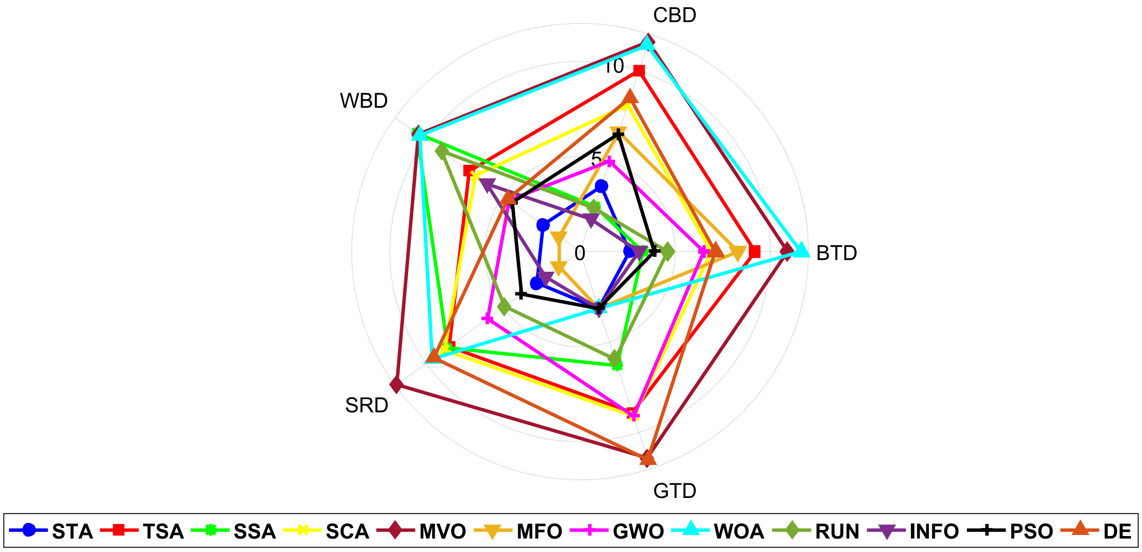


Fig. 23 Radar chart for the compared algorithms performance for real-world constrained engineering design problems

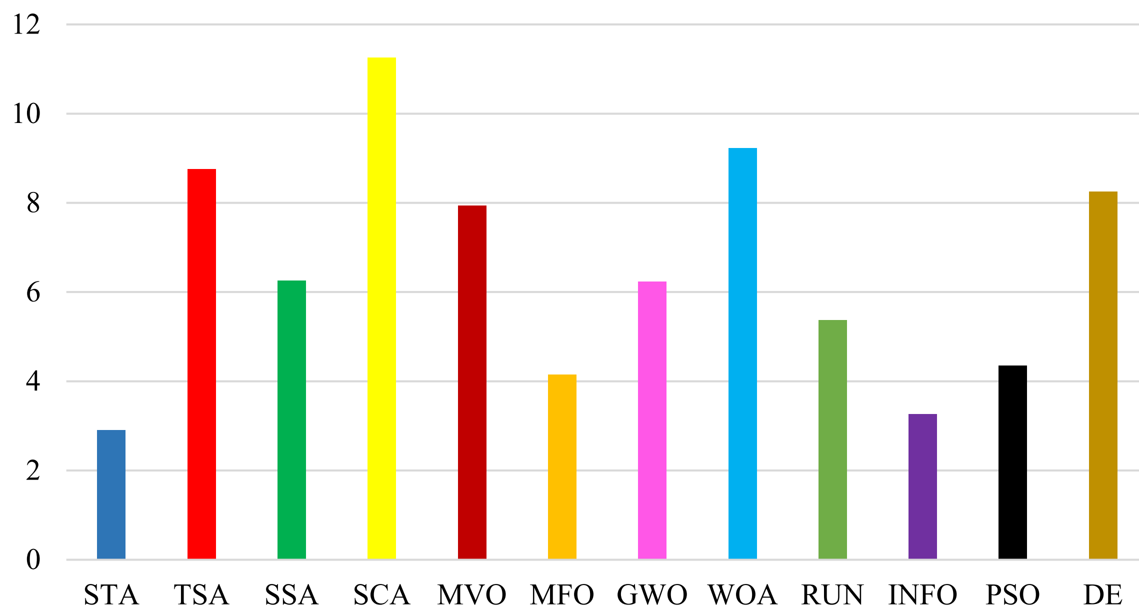


Fig. 24 Mean ranks achieved using Friedman test for real-world constrained engineering design problems

Author contributions Mohamed H. Hassan involved in conceptualization, methodology, software, writing— original draft preparation. Salah Kamel took part in conceptualization, methodology, software writing—original draft preparation.

Funding Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB).

Data and materials availability Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare that there is no conflict of interest regarding the publication of this manuscript.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent Not applicable.

References

1. Xie L, Han T, Zhou H et al (2021) Tuna swarm optimization: a novel swarm-based metaheuristic algorithm for global optimization. Comput Intell Neurosci 2021:1–22. https://doi.org/10.1155/2021/9210050
2. Alsattar HA, Zaidan AA, Zaidan BB (2020) Novel meta-heuristic bald eagle search optimisation algorithm. Artif Intell Rev 53:2237–2264. https://doi.org/10.1007/s10462-019-09732-5
3. Storn R, Price K (1997) Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. J Global Optim 11:341–359. https://doi.org/10.1023/A:1008202821328
4. Booker LB, Goldberg DE, Holland JH (1989) Classifier systems and genetic algorithms. Artif Intell 40:235–282. https://doi.org/10.1016/0004-3702(89)90050-7
5. Yao X, Liu Y, Lin G (1999) Evolutionary programming made faster. IEEE Trans Evol Comput 3:82–102. https://doi.org/10.1109/4235.771163
6. Whitley D (2001) An overview of evolutionary algorithms: practical issues and common pitfalls. Inf Softw Technol 43:817–831. https://doi.org/10.1016/S0950-5849(01)00188-4
7. Wang D, Tan D, Liu L (2018) Particle swarm optimization algorithm: an overview. Soft comput 22:387–408. https://doi.org/10.1007/s00500-016-2474-6
8. Dorigo M, Birattari M, Stutzle T (2006) Ant colony optimization. IEEE Comput Intell Mag 1:28–39. https://doi.org/10.1109/MCI.2006.329691
9. Rashedi E, Nezamabadi-pour H, Saryazdi S (2009) GSA: a gravitational search algorithm. Inf Sci (N Y) 179:2232–2248. https://doi.org/10.1016/j.ins.2009.03.004
10. Mirjalili S, Mirjalili SM, Lewis A (2014) Grey wolf optimizer. Adv Eng Softw 69:46–61. https://doi.org/10.1016/j.advengsoft.2013.12.007
11. Karaboga D, Basturk B (2007) Artificial bee colony (ABC) Optimization algorithm for solving constrained optimization problems. In: Foundations of fuzzy logic and soft computing. Springer, Berlin, Heidelberg, pp 789–798. https://doi.org/10.1007/978-3-540-72950-1_77
12. Mirjalili S (2015) Moth-flame optimization algorithm: a novel nature-inspired heuristic paradigm. Knowl Based Syst 89:228–249. https://doi.org/10.1016/j.knosys.2015.07.006
13. Mirjalili S, Lewis A (2016) The whale optimization algorithm. Adv Eng Softw 95:51–67. https://doi.org/10.1016/j.advengsoft.2016.01.008
14. Mirjalili S (2015) The ant lion optimizer. Adv Eng Softw 83:80–98. https://doi.org/10.1016/j.advengsoft.2015.01.010
15. del Valle Y, Venayagamoorthy GK, Mohagheghi S et al (2008) Particle swarm optimization: basic concepts, variants and applications in power systems. IEEE Trans Evol Comput 12:171–195. https://doi.org/10.1109/TEVC.2007.896686
16. Alamir N, Kamel S, Hassan MH, Abdelkader SM (2023) An effective quantum artificial rabbits optimizer for energy management in microgrid considering demand response. Soft comput. https://doi.org/10.1007/s00500-023-08814-5
17. Markowski P (2007) Supercell thunderstorms. In: Atmospheric convection: research and operational forecasting aspects. Springer, Vienna, pp 29–43. https://doi.org/10.1007/978-3-211-69291-2_5
18. Lemon LR (1998) On the mesocyclone "dry intrusion" and tornadogenesis. In: Preprints, 19th Conf. on severe local storms, Minneapolis, MN, Amer Meteor Soc p 755
19. Davies-Jones R (1984) Streamwise vorticity: the origin of updraft rotation in supercell storms. J Atmos Sci 41:2991–3006. https://doi.org/10.1175/1520-0469(1984)041%3c2991:SVTOOU%3e2.0.CO;2
20. Markowski P, Richardson Y (2011) Mesoscale meteorology in midlatitudes. Wiley, New Jersey
21. Faramarzi A, Heidarinejad M, Mirjalili S, Gandomi AH (2020) Marine predators algorithm: a nature-inspired metaheuristic. Expert Syst Appl 152:113377. https://doi.org/10.1016/j.eswa.2020.113377
22. Doswell CA, Bosart LF (2001) Extratropical synoptic-scale processes and severe convection. Severe convective storms. American Meteorological Society, Boston, MA, pp 27–69
23. Kaur S, Awasthi LK, Sangal AL, Dhiman G (2020) Tunicate swarm algorithm: a new bio-inspired based metaheuristic paradigm for global optimization. Eng Appl Artif Intell 90:103541. https://doi.org/10.1016/j.engappai.2020.103541
24. Mirjalili S, Gandomi AH, Mirjalili SZ et al (2017) Salp swarm algorithm: a bio-inspired optimizer for engineering design problems. Adv Eng Softw 114:163–191. https://doi.org/10.1016/j.advengsoft.2017.07.002
25. Mirjalili S (2016) SCA: a sine cosine algorithm for solving optimization problems. Knowl Based Syst 96:120–133. https://doi.org/10.1016/j.knosys.2015.12.022
26. Mirjalili S, Mirjalili SM, Hatamlou A (2016) Multi-verse optimizer: a nature-inspired algorithm for global optimization. Neural Comput Appl 27:495–513. https://doi.org/10.1007/s00521-015-1870-7

27. Ahmadianfar I, Heidari AA, Gandomi AH et al (2021) RUN beyond the metaphor: an efficient optimization algorithm based on Runge Kutta method. Expert Syst Appl 181:115079
28. Ahmadianfar I, Heidari AA, Noshadian S et al (2022) INFO: an efficient optimization algorithm based on weighted mean of vectors. Expert Syst Appl 195:116516
29. Kennedy J, Eberhart R Particle swarm optimization. In: Proceedings of ICNN'95—international conference on neural networks. IEEE, pp 1942–1948
30. Wang Y, Wang P, Zhang J et al (2019) A novel bat algorithm with multiple strategies coupling for numerical optimization. Mathematics 7:135. https://doi.org/10.3390/math7020135
31. Sang-To T, Le-Minh H, Mirjalili S et al (2022) A new movement strategy of grey wolf optimizer for optimization problems and structural damage identification. Adv Eng Softw 173:103276. https://doi.org/10.1016/j.advengsoft.2022.103276
32. Yang Y, Gao Y, Tan S et al (2022) An opposition learning and spiral modelling based arithmetic optimization algorithm for global continuous optimization problems. Eng Appl Artif Intell 113:104981. https://doi.org/10.1016/j.engappai.2022.104981
33. Ray T, Saini P (2001) Engineering design optimization using a swarm with an intelligent information sharing among individuals. Eng Optim 33:735–748. https://doi.org/10.1080/03052150108940941
34. Erdogan Yildirim A, Karci A (2018) Application of three bar truss problem among engineering design optimization problems using artificial atom algorithm. In: 2018 international conference on artificial intelligence and data processing (IDAP). IEEE, pp 1–5
35. Sadollah A, Bahreininejad A, Eskandar H, Hamdi M (2013) Mine blast algorithm: a new population based algorithm for solving constrained engineering optimization problems. Appl Soft Comput 13:2592–2612. https://doi.org/10.1016/j.asoc.2012.11.026
36. Zhang M, Luo W, Wang X (2008) Differential evolution with dynamic stochastic selection for constrained optimization. Inf Sci (N Y) 178:3043–3074. https://doi.org/10.1016/j.ins.2008.02.014
37. Saremi S, Mirjalili S, Lewis A (2017) Grasshopper optimisation algorithm: theory and application. Adv Eng Softw 105:30–47. https://doi.org/10.1016/j.advengsoft.2017.01.004
38. Liu H, Cai Z, Wang Y (2010) Hybridizing particle swarm optimization with differential evolution for constrained numerical and engineering optimization. Appl Soft Comput 10:629–640. https://doi.org/10.1016/j.asoc.2009.08.031
39. Gandomi AH, Yang X-S, Alavi AH (2013) Cuckoo search algorithm: a metaheuristic approach to solve structural optimization problems. Eng Comput 29:17–35. https://doi.org/10.1007/s00366-011-0241-y
40. Fan Q, Huang H, Li Y et al (2021) Beetle antenna strategy based grey wolf optimization. Expert Syst Appl 165:113882. https://doi.org/10.1016/j.eswa.2020.113882
41. Chickermane H, Gea HC (1996) Structural optimization using a new local approximation method. Int J Numer Method Eng 39:829–846
42. Cheng M-Y, Prayogo D (2014) Symbiotic organisms search: a new metaheuristic optimization algorithm. Comput Struct 139:98–112. https://doi.org/10.1016/j.compstruc.2014.03.007
43. Zhao J, Gao Z-M, Sun W (2020) The improved slime mould algorithm with Levy flight. J Phys Conf Ser 1617:012033. https://doi.org/10.1088/1742-6596/1617/1/012033
44. Yu C, Cai Z, Ye X et al (2020) Quantum-like mutation-induced dragonfly-inspired optimization approach. Math Comput Simul 178:259–289. https://doi.org/10.1016/j.matcom.2020.06.012
45. Kaveh A, Khayatazad M (2012) A new meta-heuristic method: ray optimization. Comput Struct 112–113:283–294. https://doi.org/10.1016/j.compstruc.2012.09.003
46. Huang F, Wang L, He Q (2007) An effective co-evolutionary differential evolution for constrained optimization. Appl Math Comput 186:340–356. https://doi.org/10.1016/j.amc.2006.07.105
47. Deb K (1991) Optimal design of a welded beam via genetic algorithms. AIAA J 29:2013–2015. https://doi.org/10.2514/3.10834
48. Lee KS, Geem ZW (2005) A new meta-heuristic algorithm for continuous engineering optimization: harmony search theory and practice. Comput Method Appl Mech Eng 194:3902–3933. https://doi.org/10.1016/j.cma.2004.09.007
49. He Q, Wang L (2007) An effective co-evolutionary particle swarm optimization for constrained engineering design problems. Eng Appl Artif Intell 20:89–99. https://doi.org/10.1016/j.engappai.2006.03.003
50. Ghasemi M, Rahimnejad A, Akbari E et al (2023) A new metaphor-less simple algorithm based on Rao algorithms: a fully informed search algorithm (FISA). PeerJ Comput Sci 9:e1431. https://doi.org/10.7717/peerj-cs.1431
51. Seyyedabbasi A, Kiani F (2023) Sand cat swarm optimization: a nature-inspired algorithm to solve global optimization problems. Eng Comput 39:2627–2651. https://doi.org/10.1007/s00366-022-01604-x
52. Mezura-Montes E, Coello Coello CA, Landa-Becerra R Engineering optimization using simple evolutionary algorithm. In: Proceedings. 15th IEEE international conference on tools with artificial intelligence. IEEE Comput. Soc, pp 149–156
53. Stephen S, Christu D, Dalvi A (2018) Design optimization of weight of speed reducer problem through matlab and simulation using ansys. Int J Mech Eng Technol 9:339–349
54. Kamboj VK, Nandi A, Bhadoria A, Sehgal S (2020) An intensify Harris Hawks optimizer for numerical and engineering optimization problems. Appl Soft Comput 89:106018. https://doi.org/10.1016/j.asoc.2019.106018
55. Lu S, Kim HM (2010) A regularized inexact penalty decomposition algorithm for multidisciplinary design optimization problems with complementarity constraints. J Mech Des 132:041005. https://doi.org/10.1115/1.4001206
56. Geem ZW, Kim JH, Loganathan GV (2001) A new heuristic optimization algorithm: harmony search. SIMULATION 76:60–68
57. Baykasoğlu A, Ozsoydan FB (2015) Adaptive firefly algorithm with chaos for mechanical design optimization problems. Appl Soft Comput 36:152–164. https://doi.org/10.1016/j.asoc.2015.06.056
58. Akhtar S, Tai K, Ray T (2002) A socio-behavioural simulation model for engineering design optimization. Eng Optim 34:341–354. https://doi.org/10.1080/03052150212723
59. Abualigah L, Yousri D, Abd Elaziz M et al (2021) Aquila optimizer: a novel meta-heuristic optimization algorithm. Comput Ind Eng 157:107250. https://doi.org/10.1016/j.cie.2021.107250
60. Ma B, Lu P, Zhang L et al (2021) Enhanced sparrow search algorithm with mutation strategy for global optimization. IEEE Access 9:159218–159261. https://doi.org/10.1109/ACCESS.2021.3129255